

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

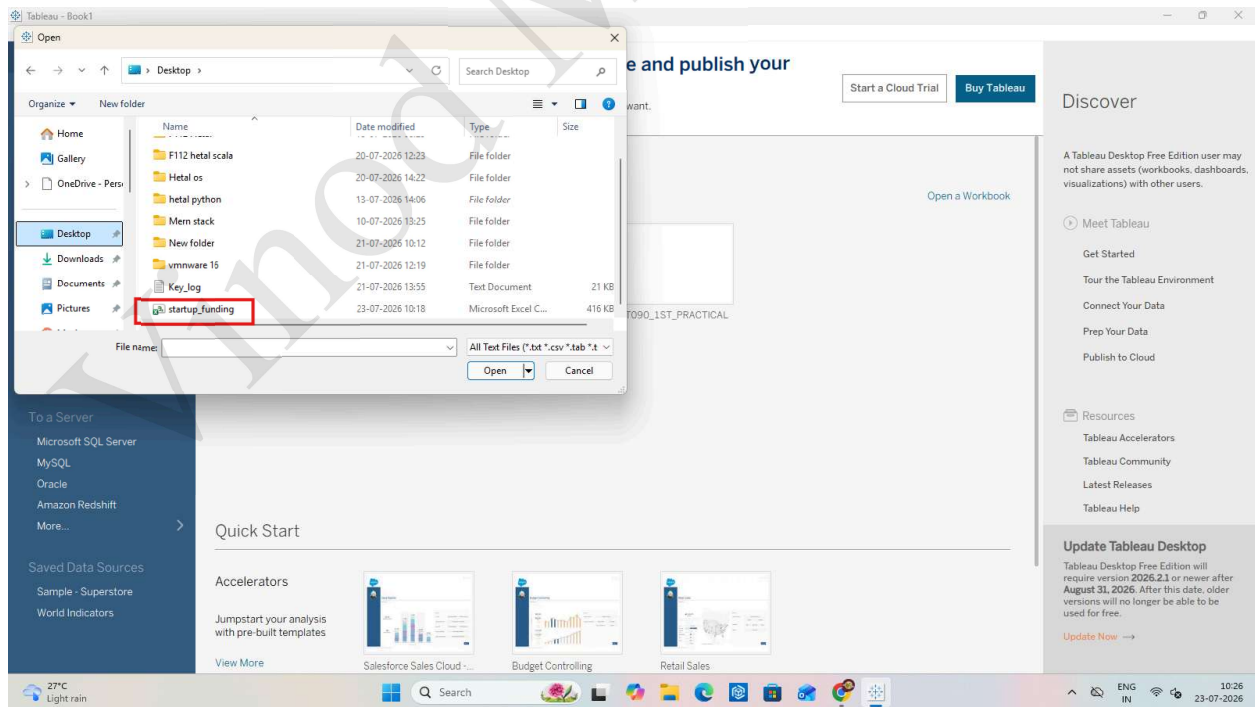
Aim :- Creating Advanced Visualizations in Tableau

- a) Create stacked and grouped bar charts for comparison
- b) Build treemaps and pie charts for part-to-whole analysis
- c) Design scatter plots to identify relationships
- d) Create dual-axis and combination charts
- e) Develop histograms for distribution analysis
- f) Build box-and-whisker plots
- g) Use Gantt charts and advanced date visualizations

Step A: Creating Stacked and Grouped Bar Charts for Comparison

1. Connecting to the Dataset

1. Launch Tableau. Under the Connect panel, click on Text File, select your startup_funding.csv, and click Open.

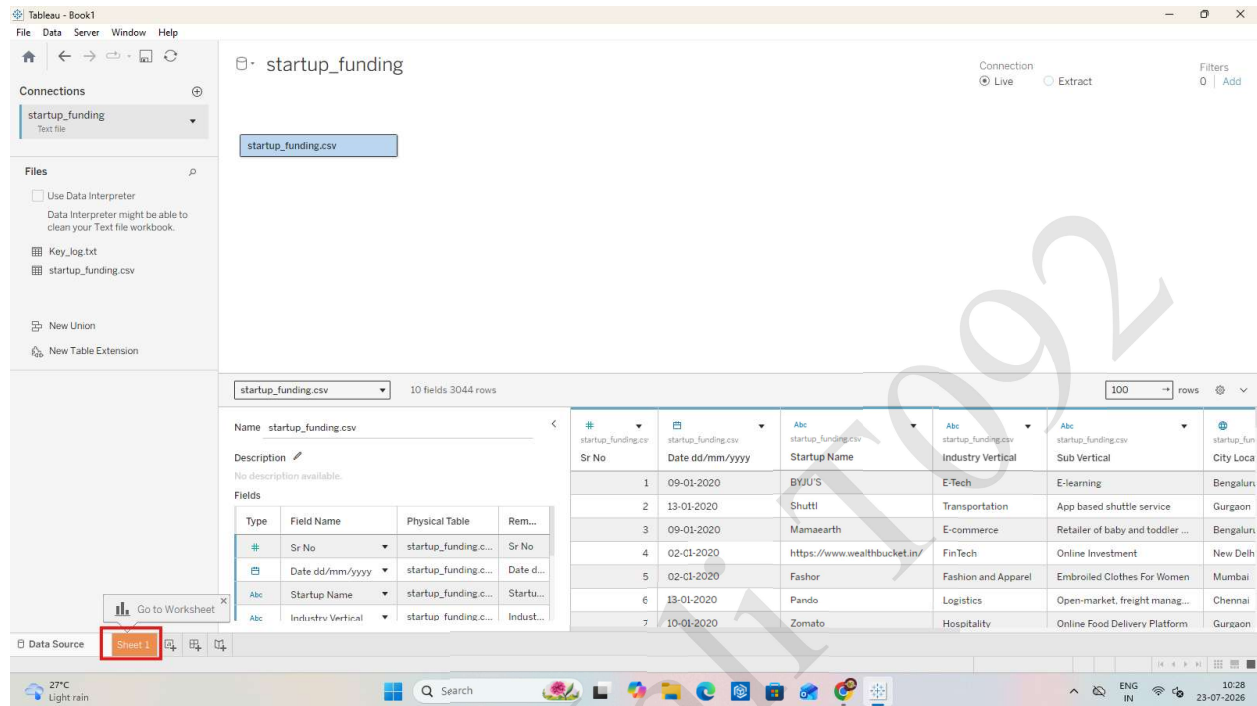


Name :- Vinod Manohar Mali

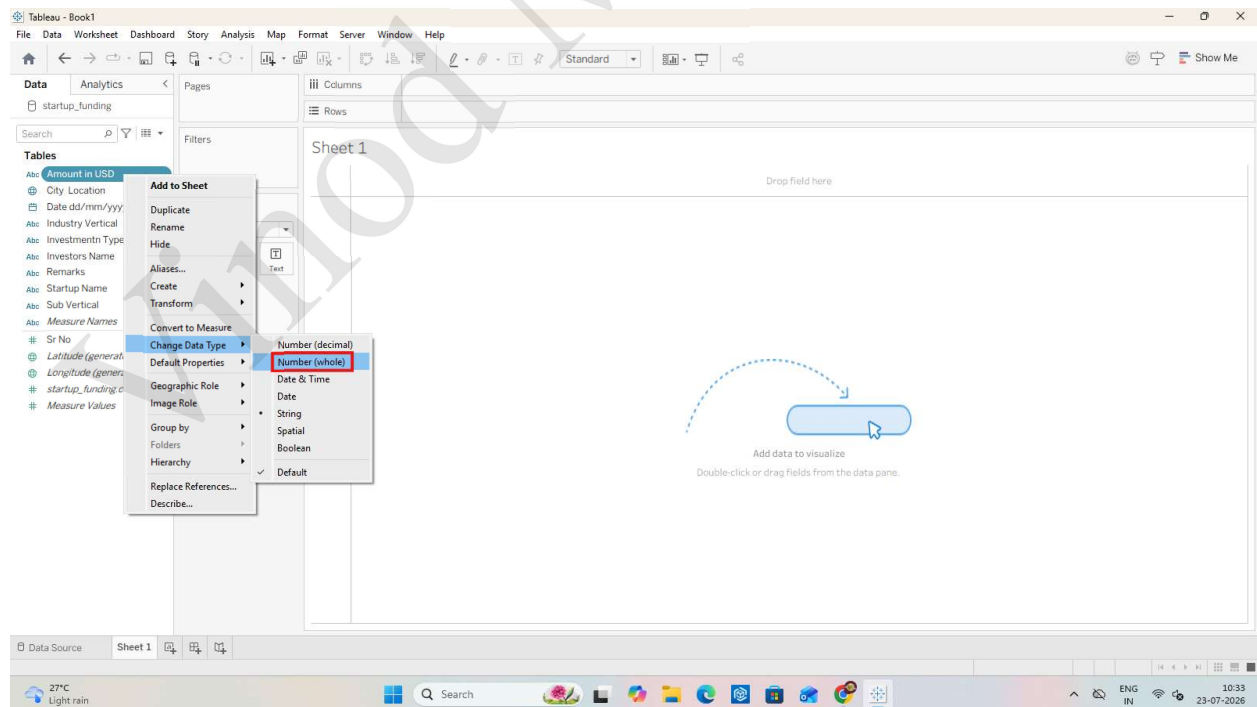
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

2. Click on Sheet 1 at the bottom-left corner to enter your workspace.



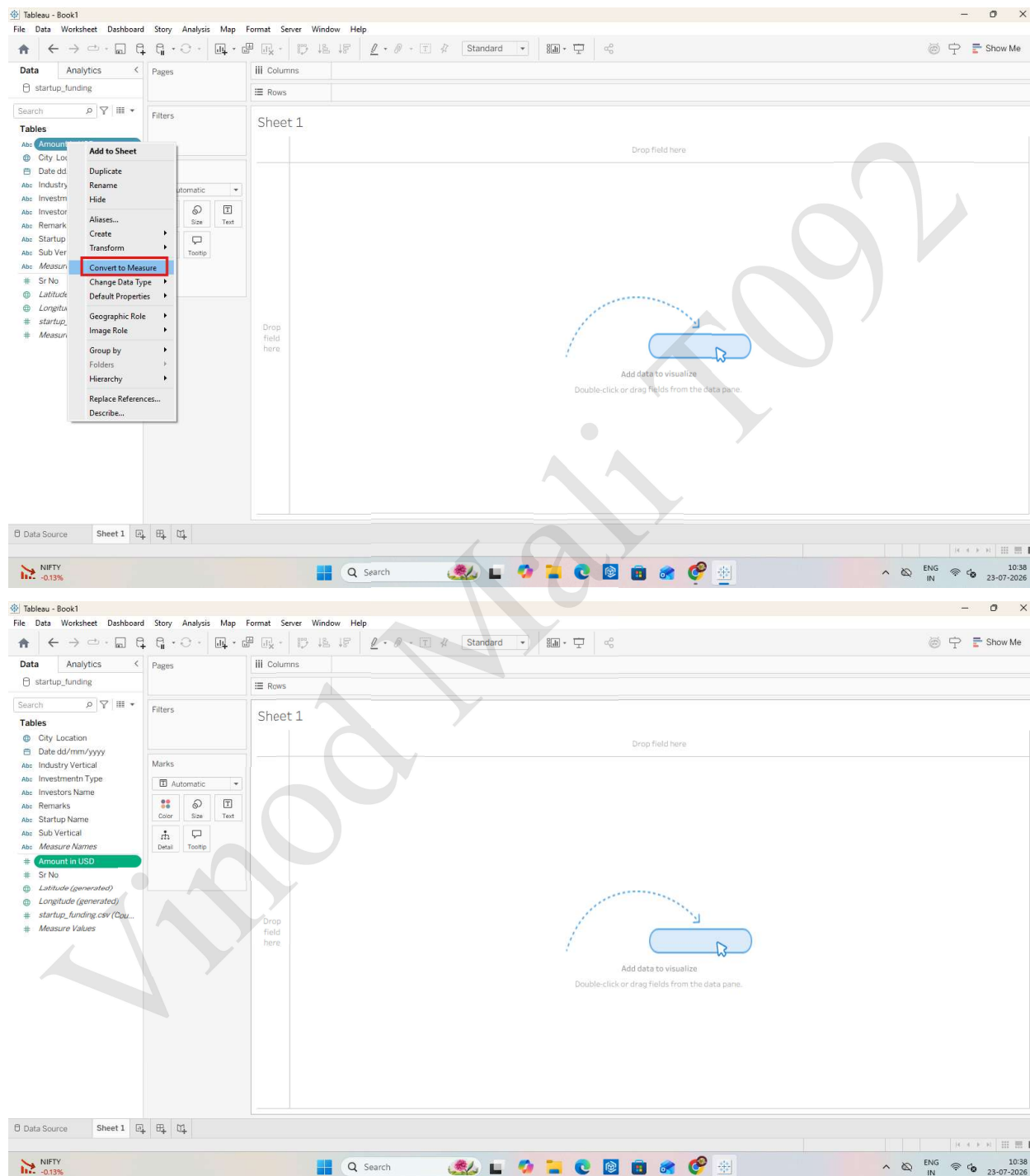
3. **Fix the Data Type:** In the Data pane on the left, locate **Amount in USD**. Click the small **Abc** text icon next to it and select **Number (Whole)** from the dropdown menu to convert it into numeric data.



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

4. Convert to Measure: Right-click **Amount in USD** in the Data pane and select **Convert to Measure**. It will turn green and move to the bottom half of the pane.

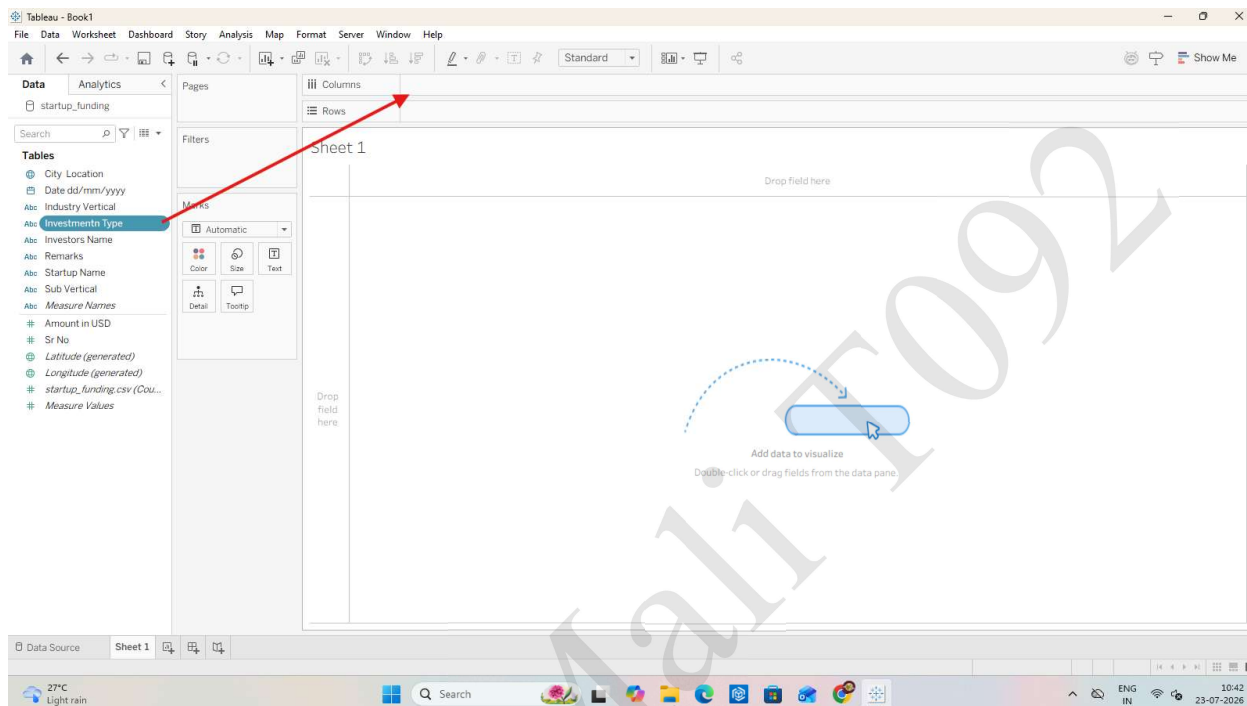


Name :- Vinod Manohar Mali
Roll no :- T092

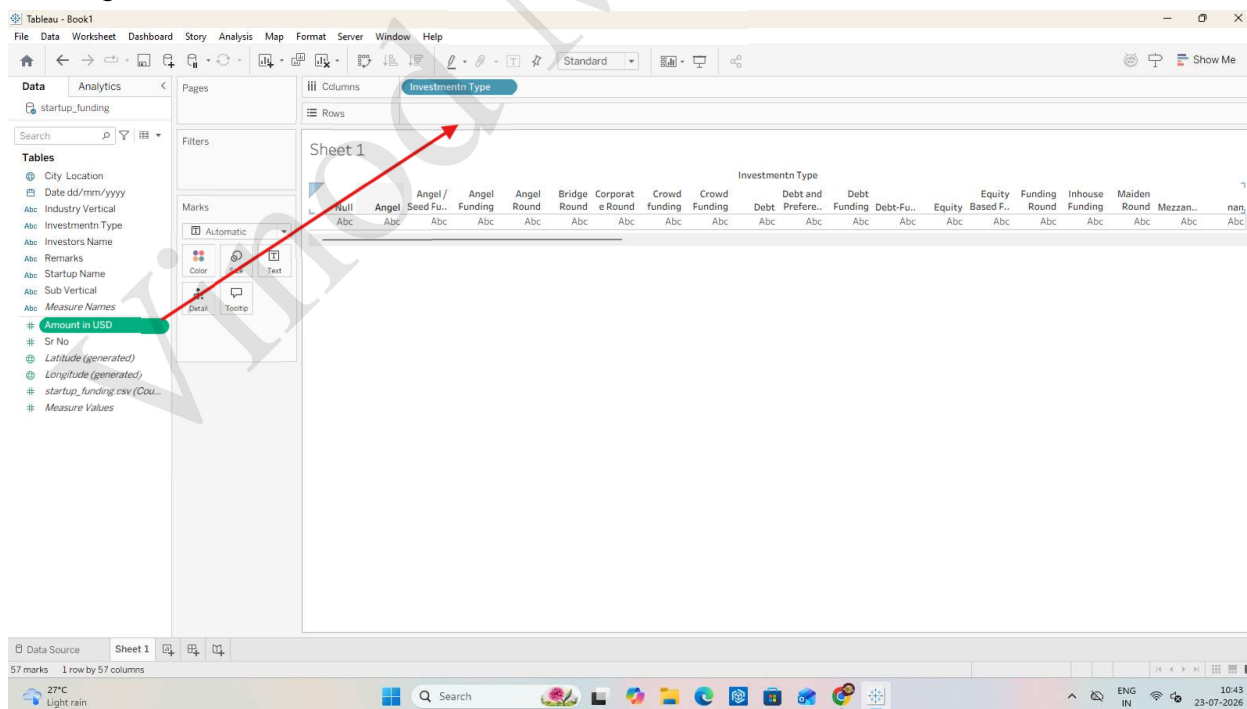
Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

2 :- Building a Stacked Bar Chart

1. From the Data Pane on the left, click and drag the **Investment Type** dimension onto the **Columns** shelf.



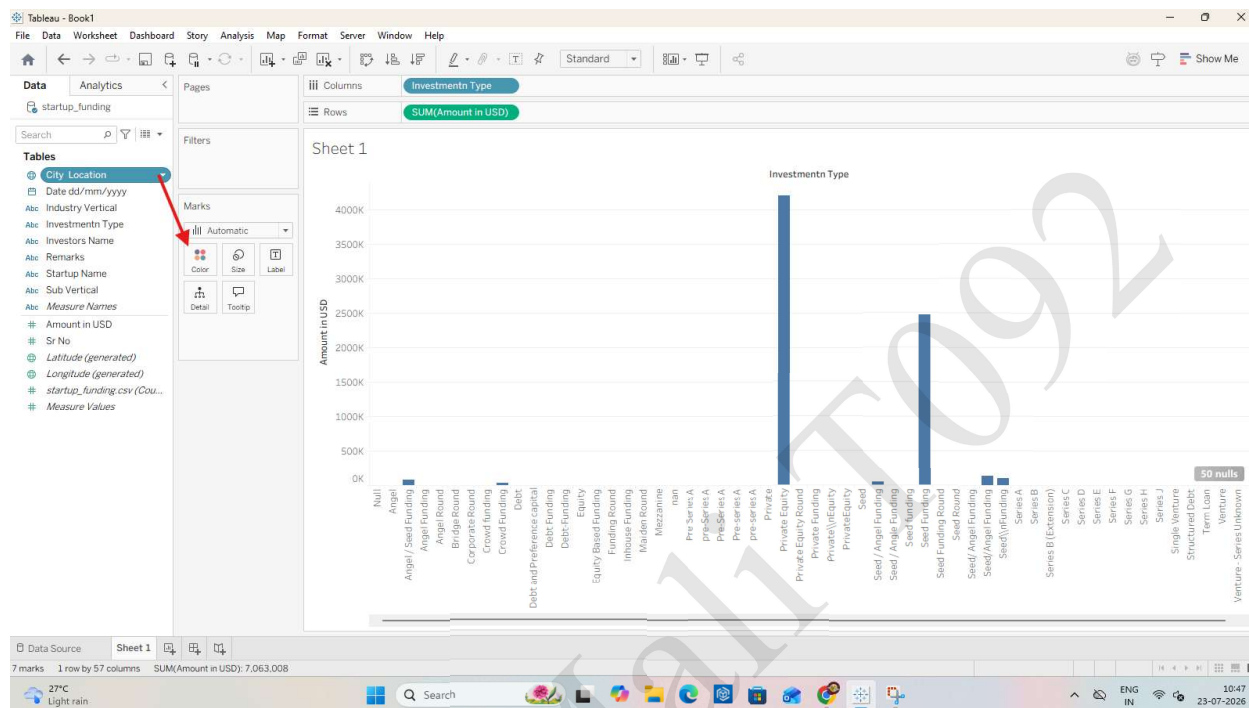
2. Drag the **Amount in USD** measure onto the Rows shelf.



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

3. Take the **City Location** dimension from the Data Pane and drop it directly onto the **Color** tile in the Marks card.

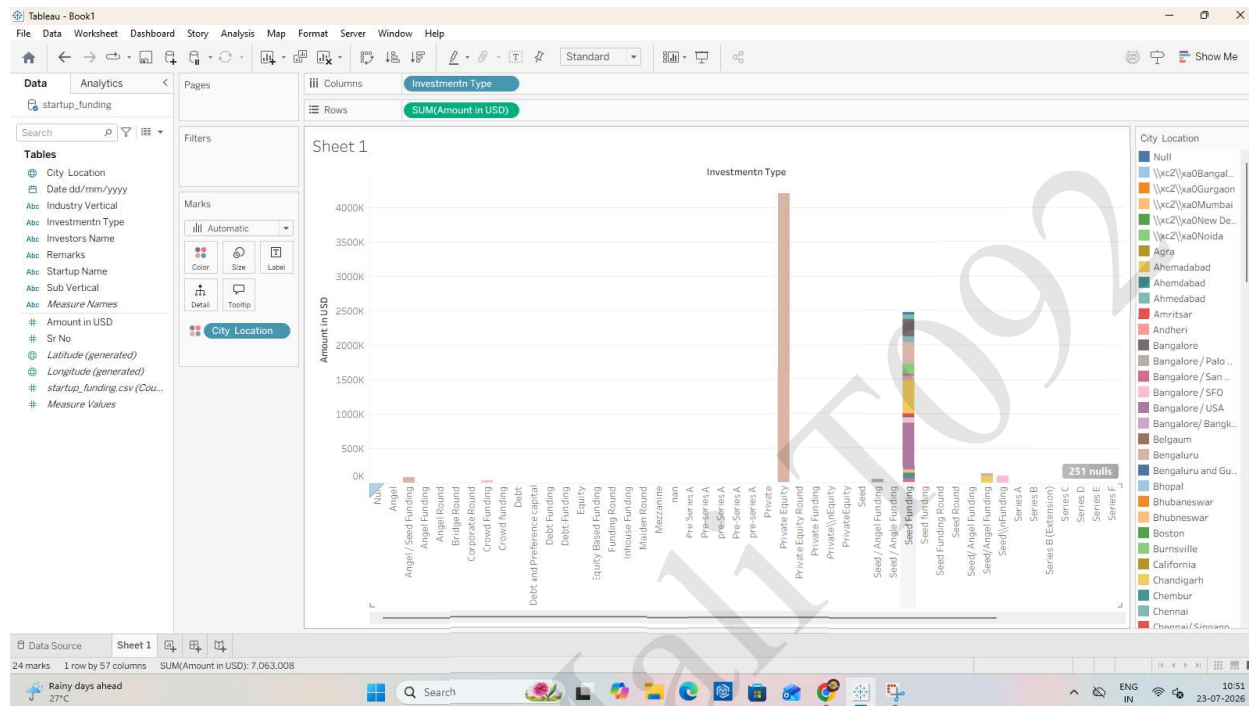


Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

3 :- Converting to a Grouped Bar Chart (Side-by-Side)

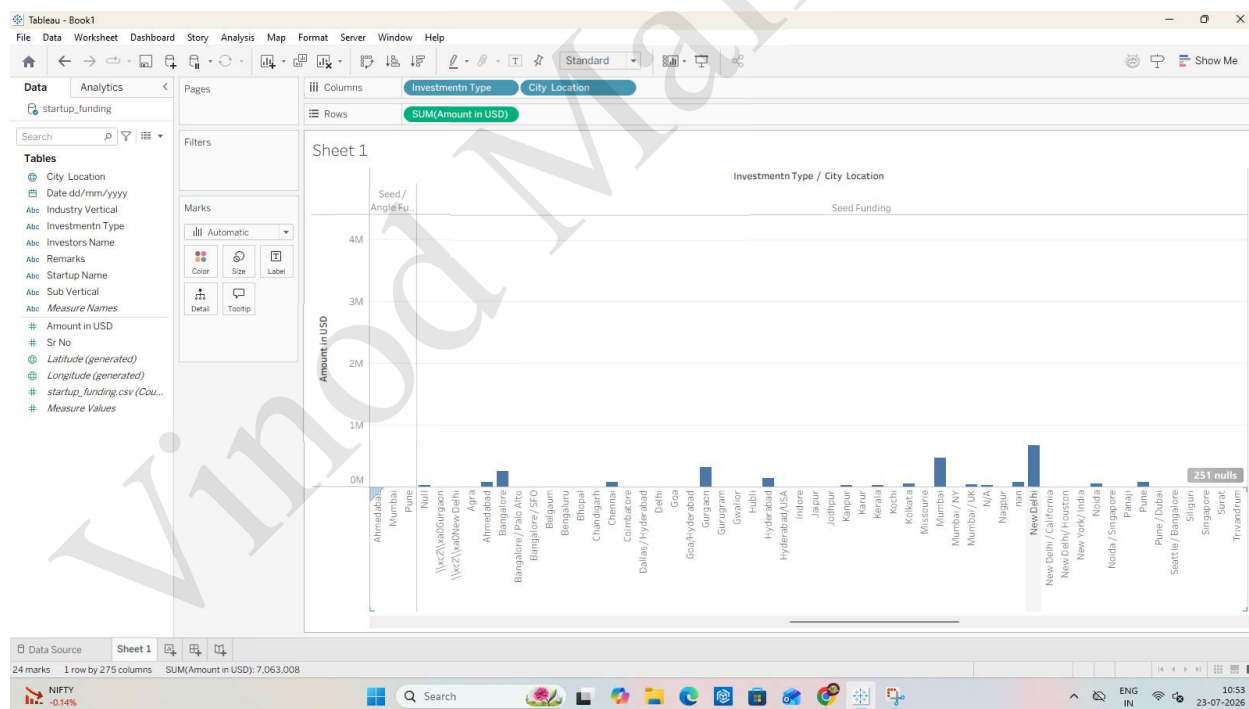
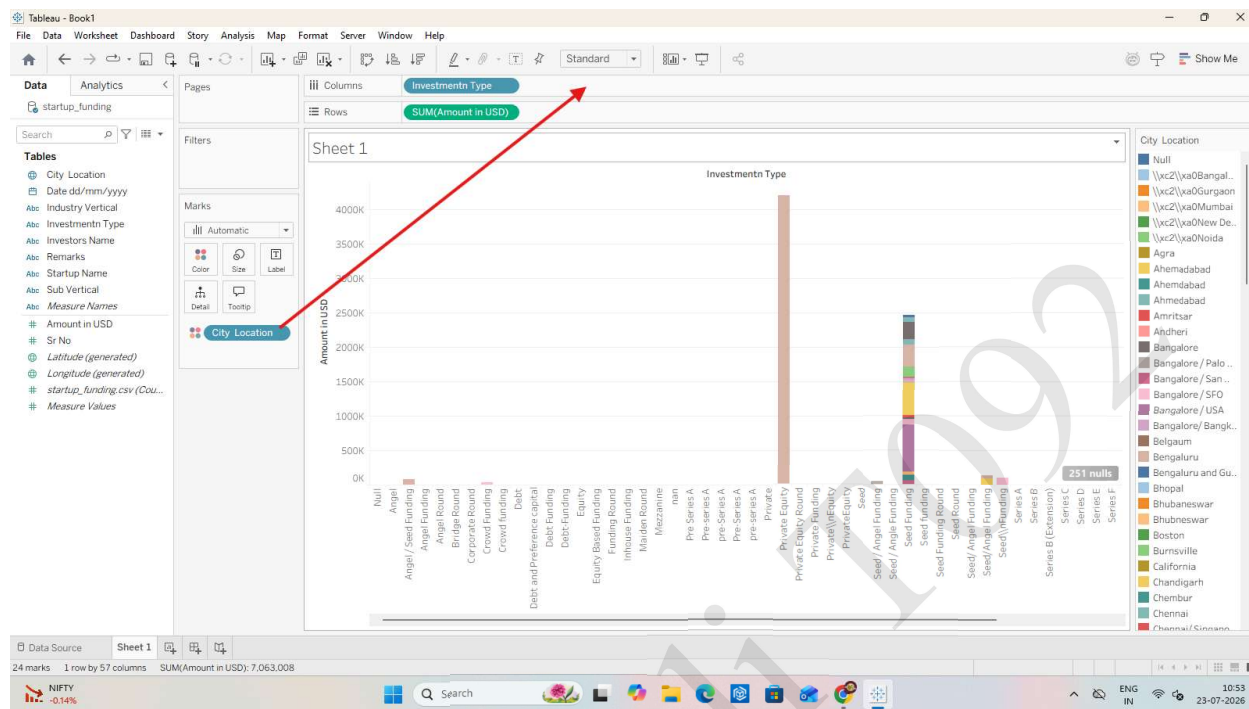
1. Leave Amount in USD on the Rows shelf.



2. Take the **City Location** pill out of the Color tile on the Marks card and drag it up to the **Columns** shelf, placing it right next to (to the left of) the **InvestmentnType** pill.

Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3



Step B: Building Treemaps and Pie Charts for Part-to-Whole Analysis

1. Building a Treemap

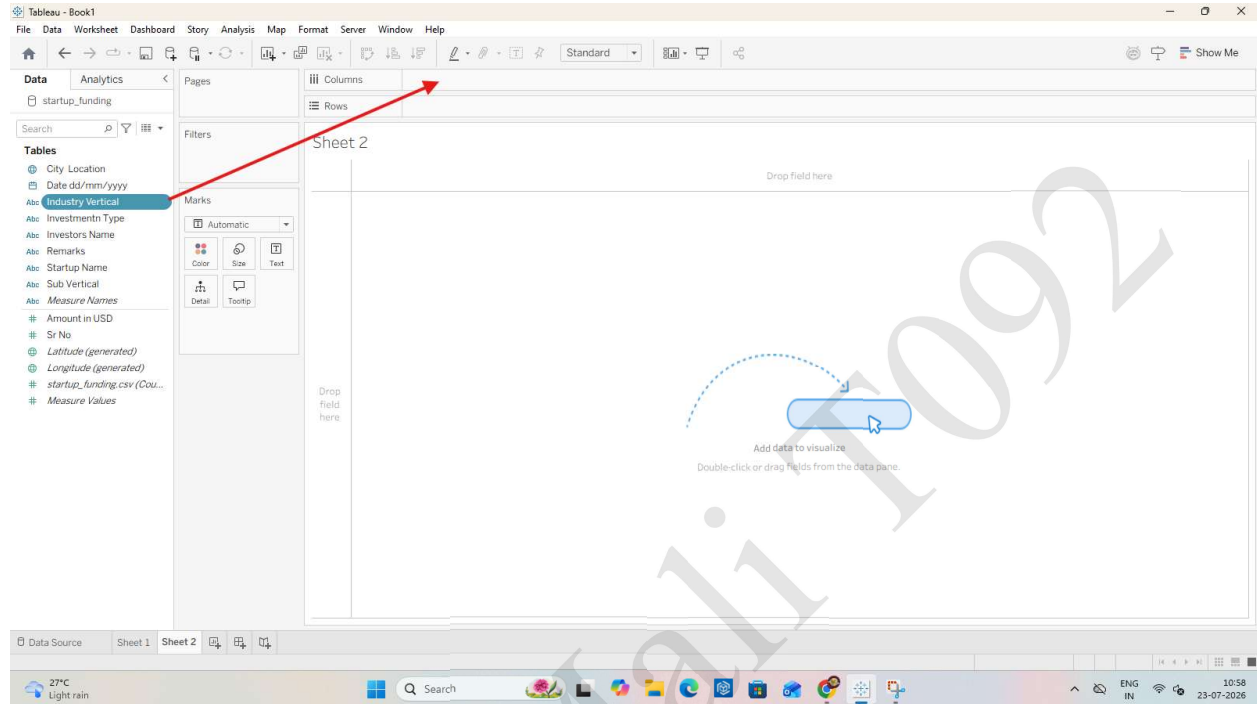
1. Click the **New Worksheet** icon at the bottom of the screen to open a clean workspace (**Sheet 2**).

Name :- Vinod Manohar Mali

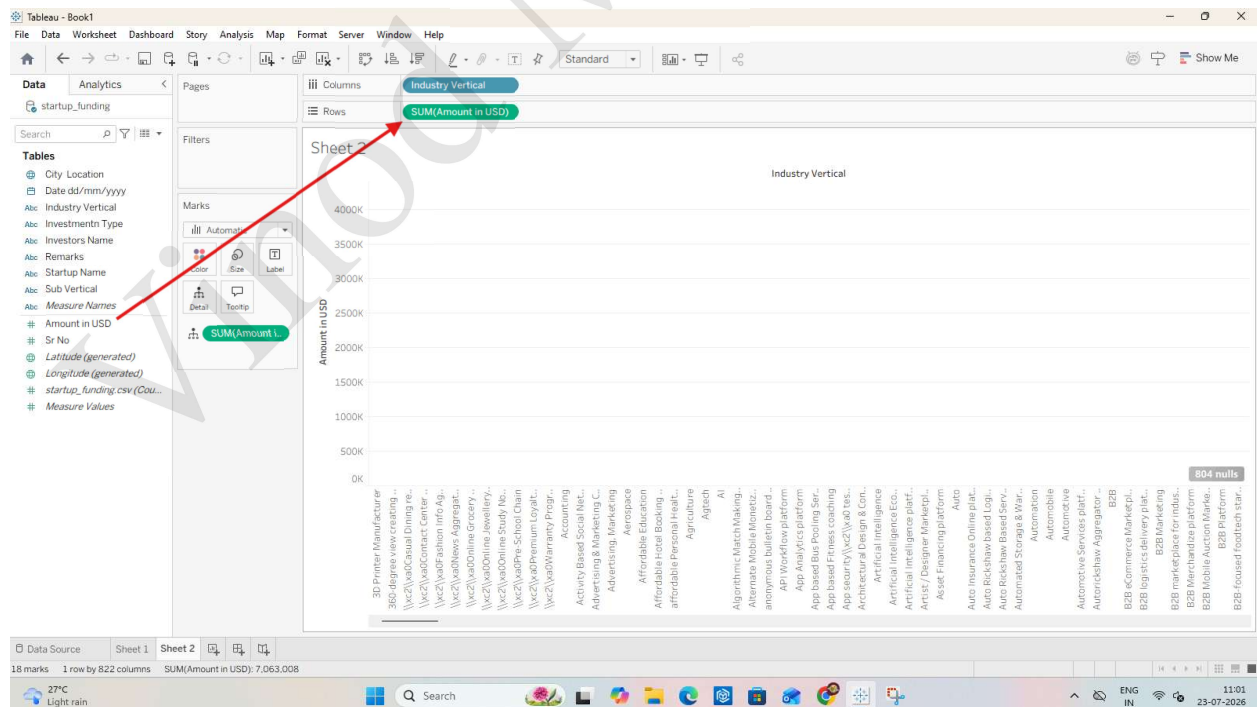
Roll no :- T092

Practical no 3

2. From the Data Pane, drag the Industry Vertical dimension and drop it onto the **Columns** shelf.



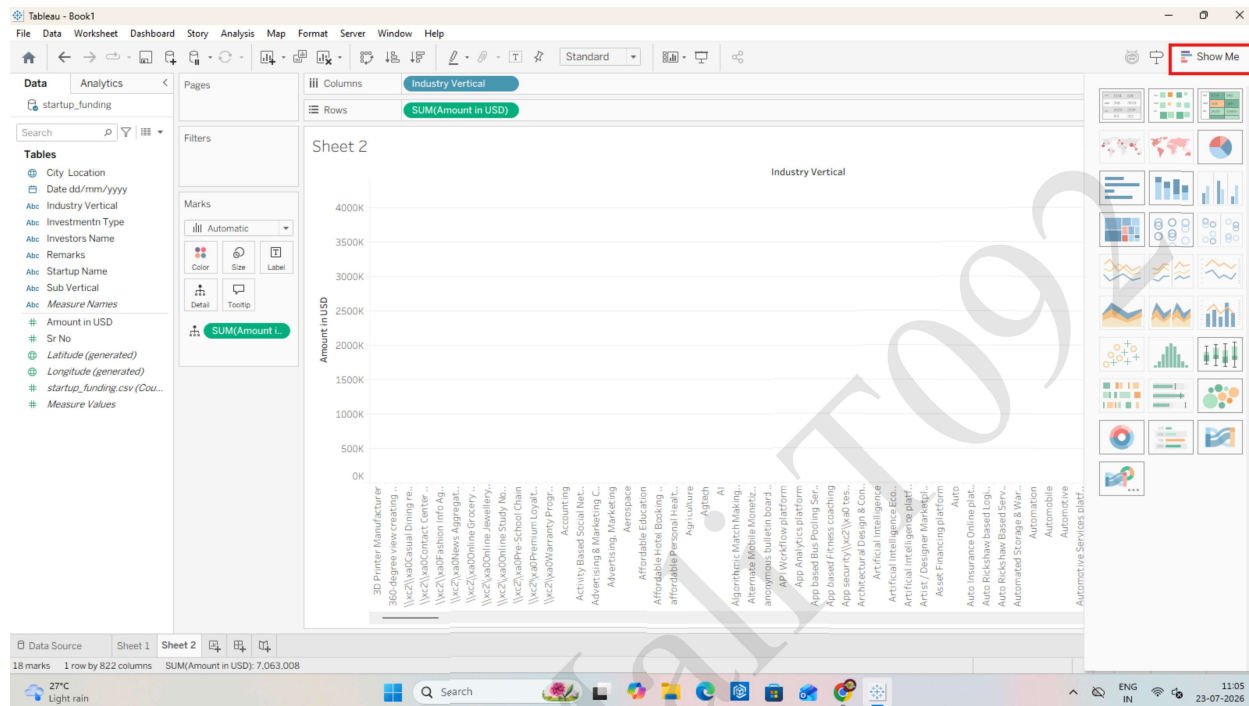
3. Drag the Amount in USD measure and drop it onto the **Rows** shelf. (Tableau will default to generating a standard vertical bar chart).



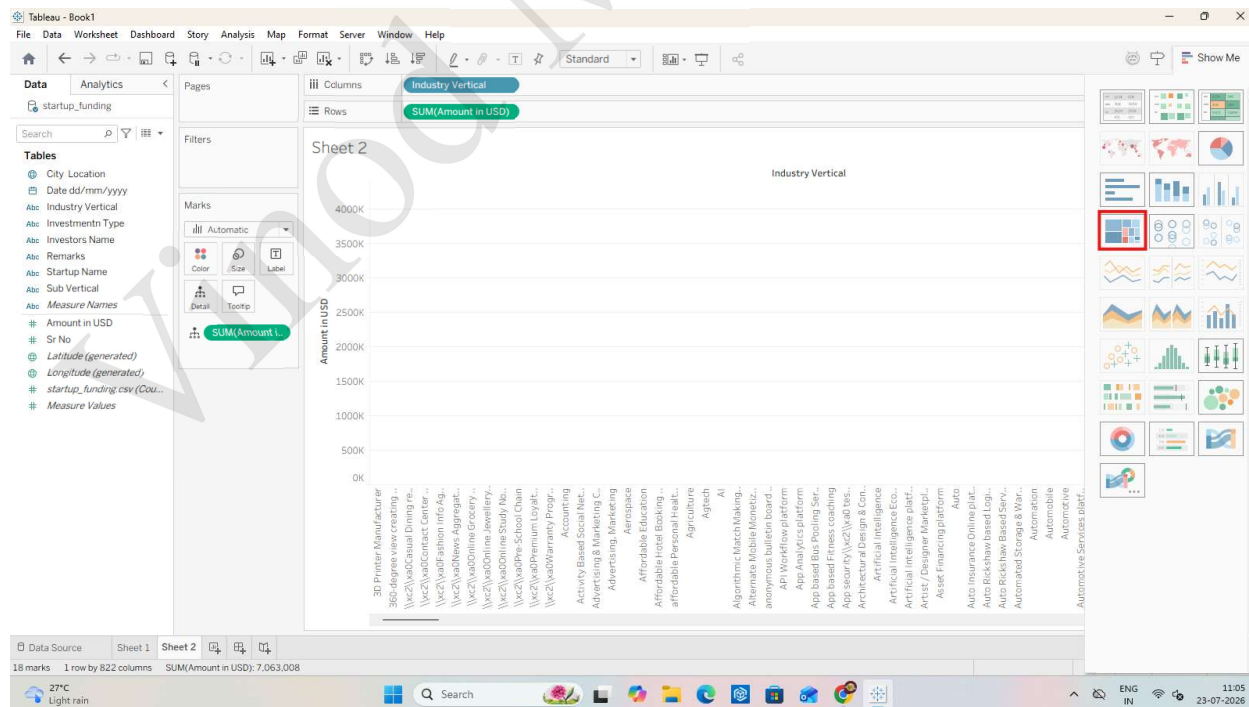
Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

4. Now, look at the very top-right corner of the window and click the **Show Me** button to expand the chart library panel.

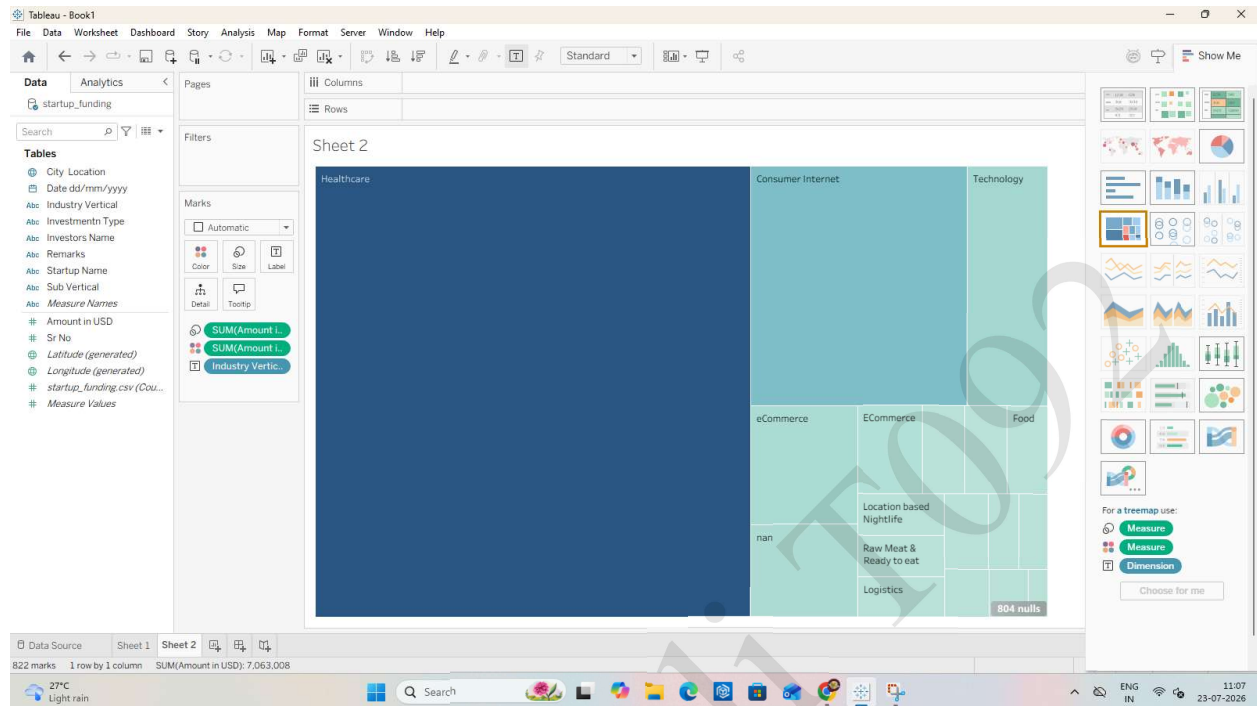


5. From the options, click on the Treemap icon (it looks like a grid of nested colorful rectangles).



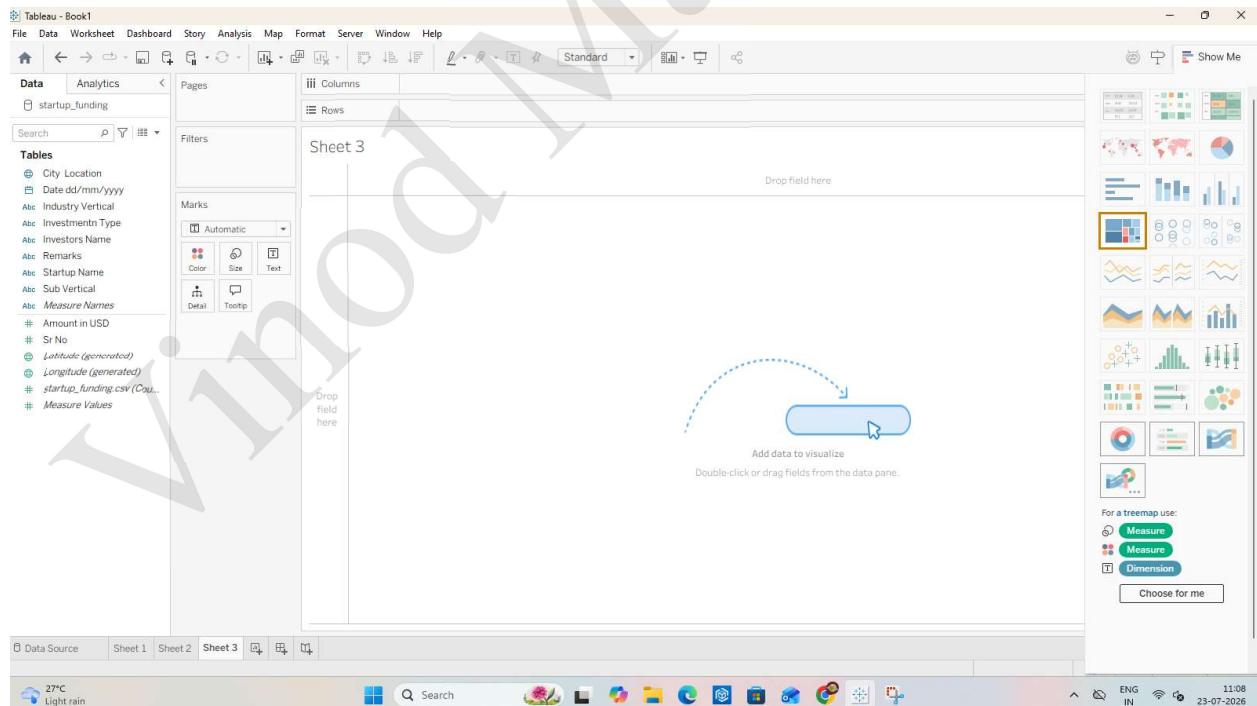
Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3



2. Building a Pie Chart

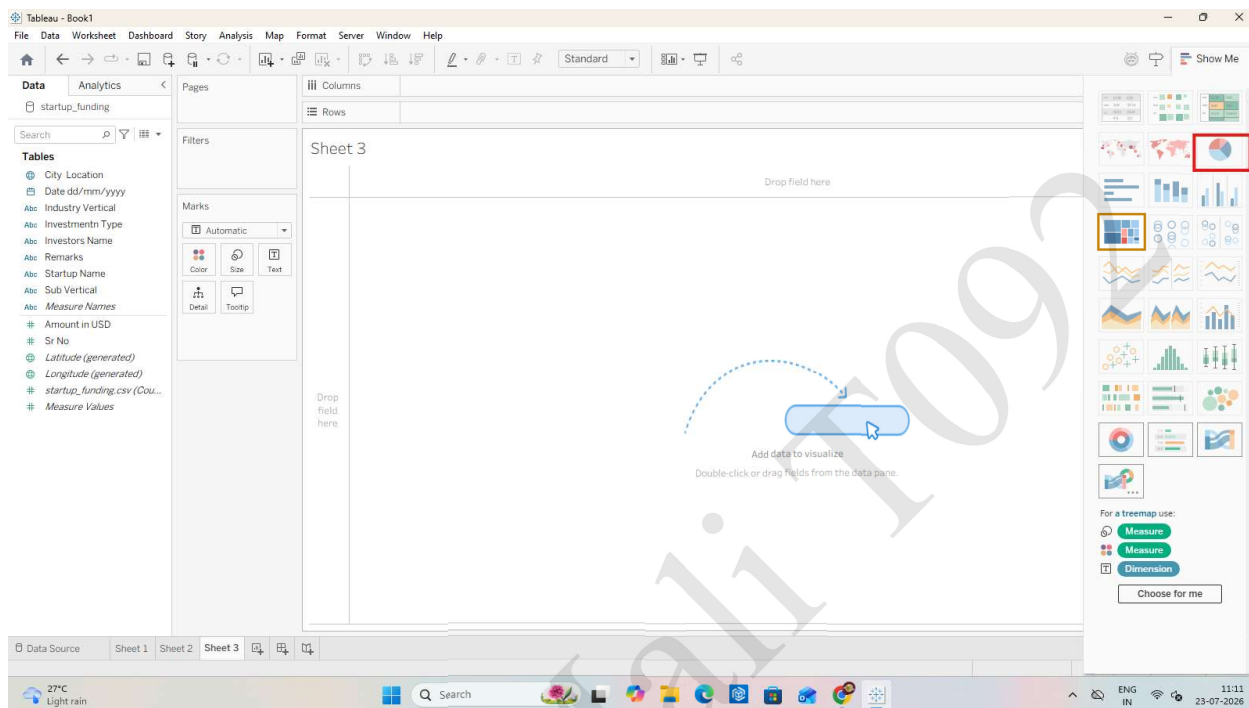
1. Click the New Worksheet icon to open another blank layout (Sheet 3).



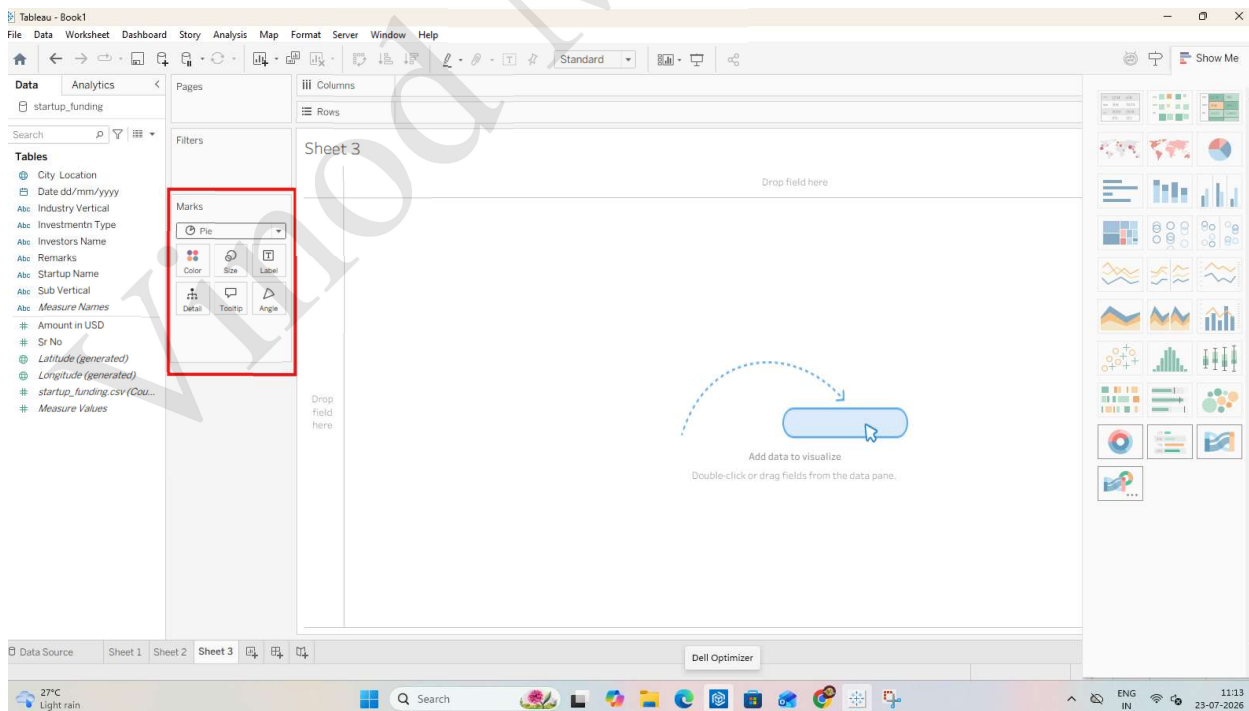
Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

2. Go back to the **Show Me** panel in the top-right corner and select the Pie Chart icon.

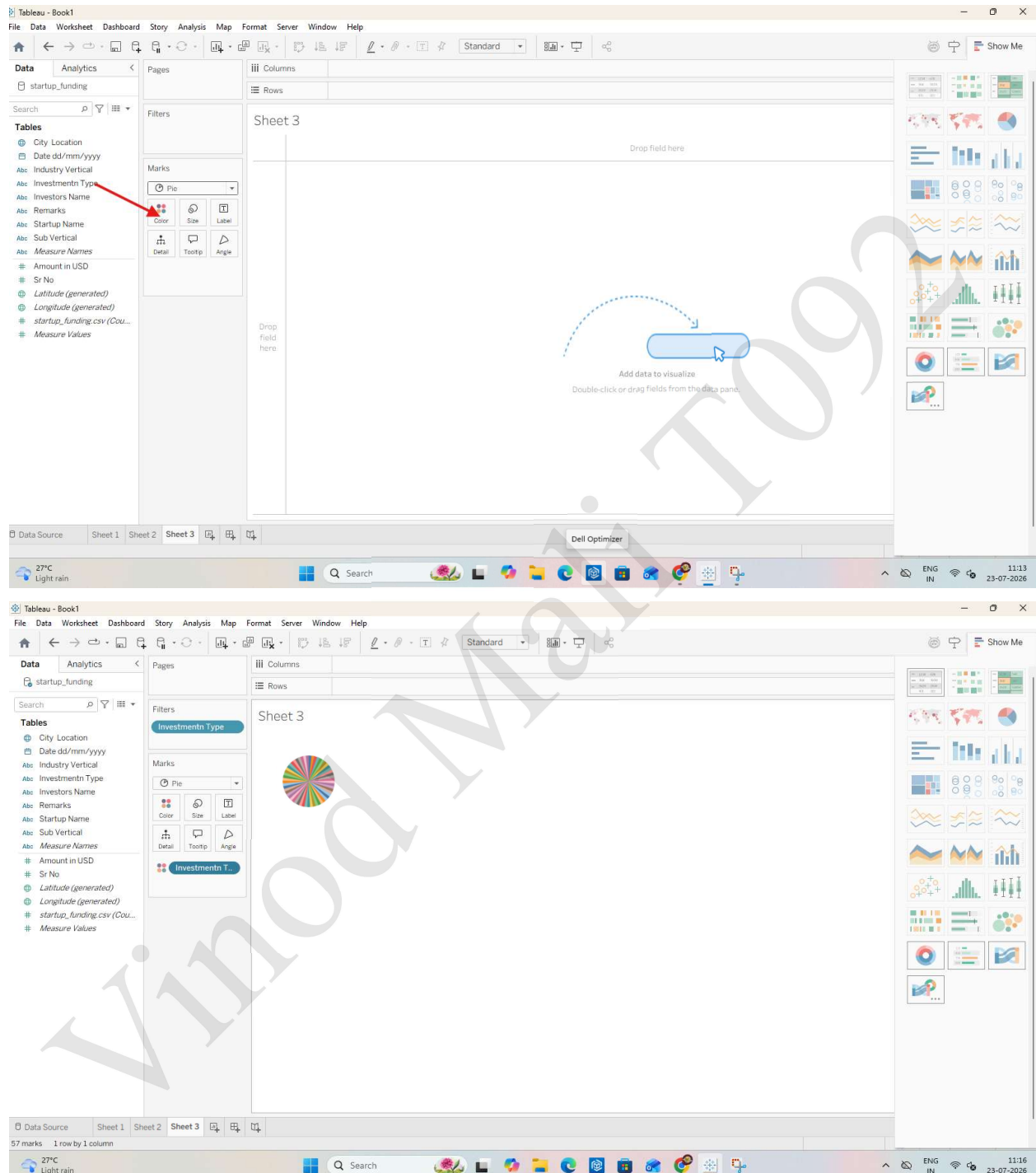


3. Notice that your Marks card changes to show an angle option. From the Data Pane, drag InvestmentType and drop it onto the Color tile in the Marks card.



Name :- Vinod Manohar Mali
Roll no :- T092

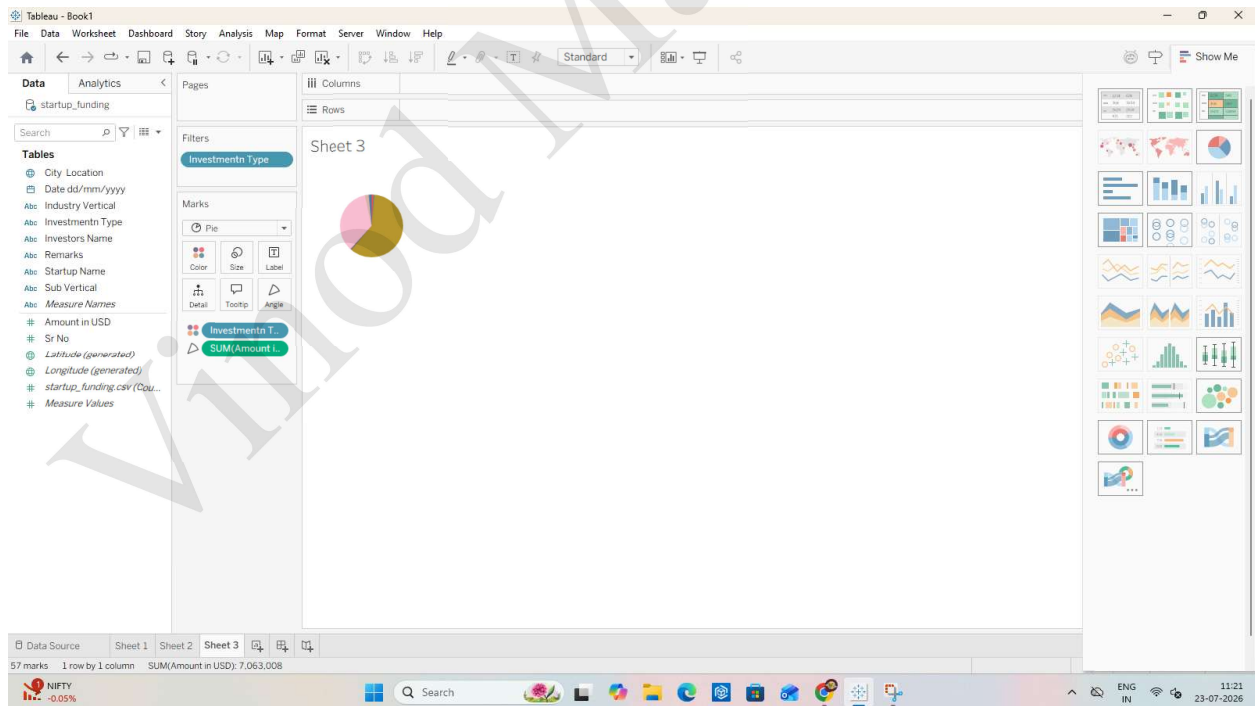
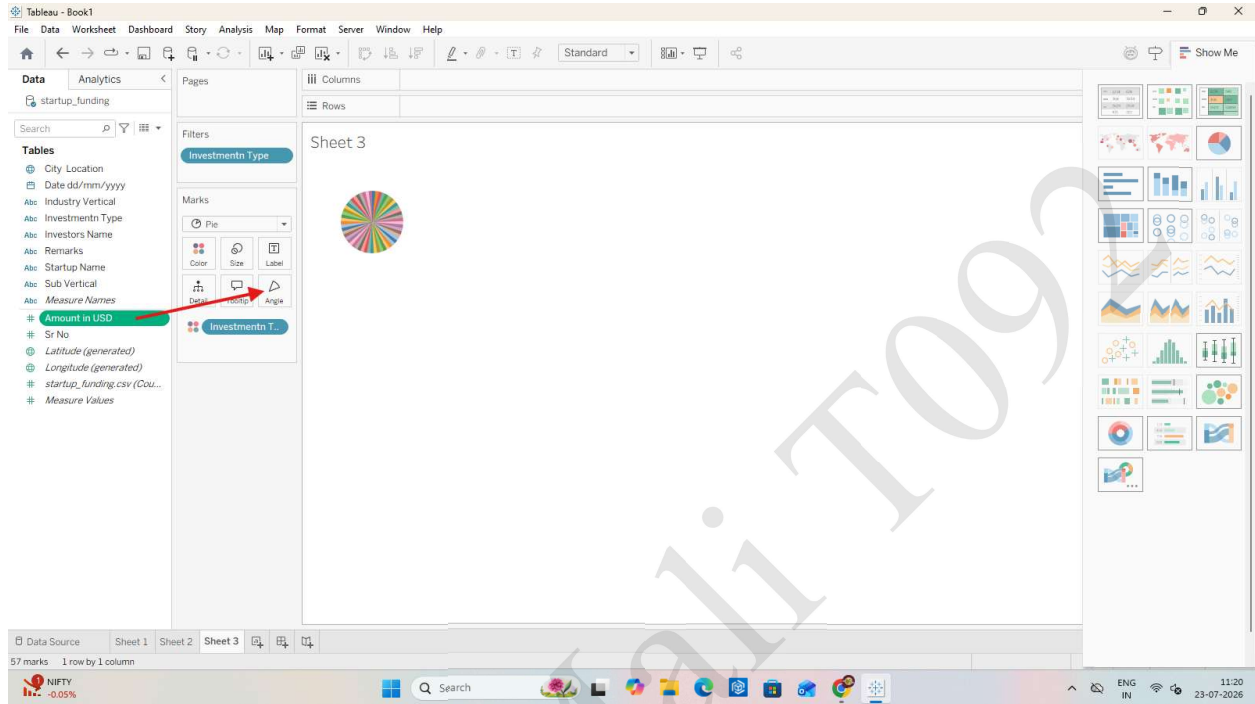
Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

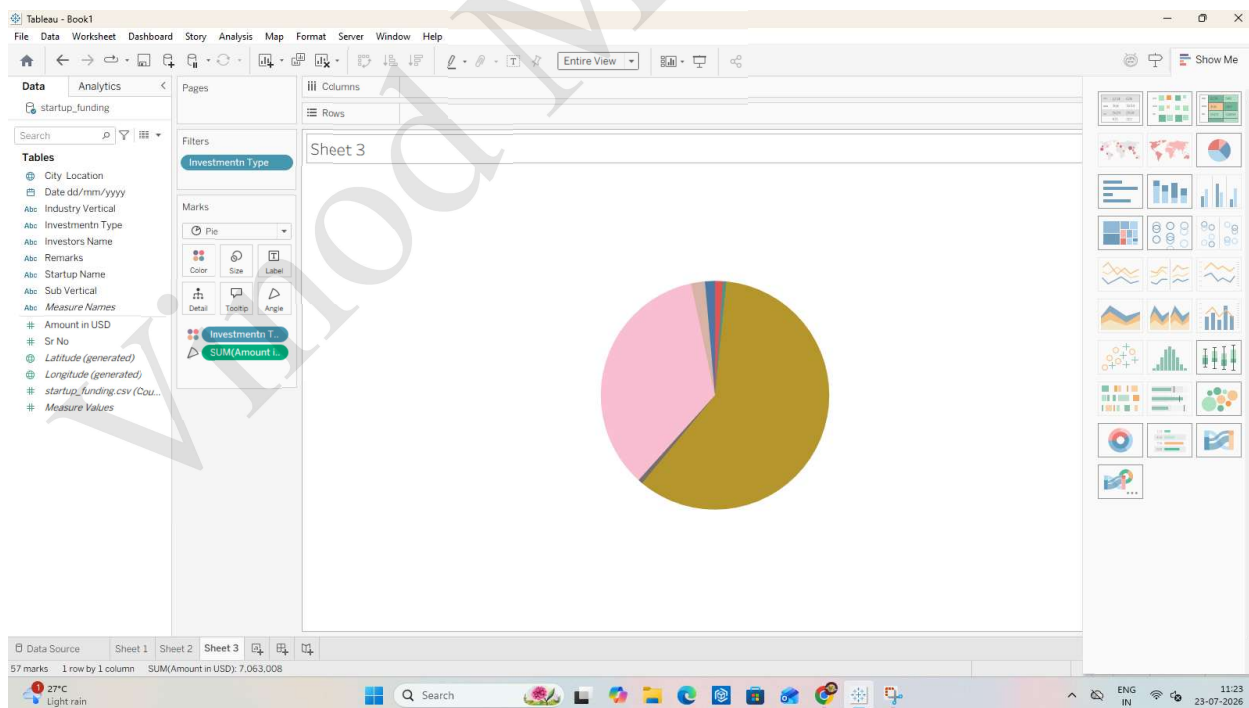
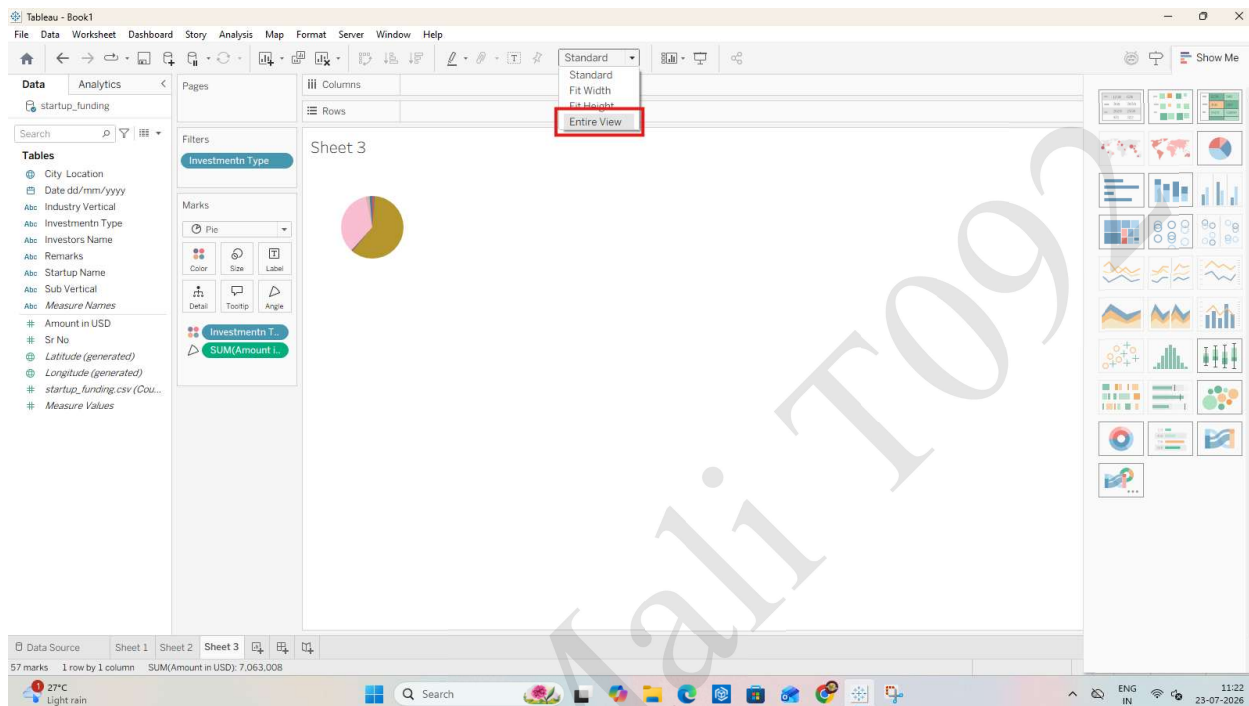
4. Drag Amount in USD from the Data Pane and drop it directly onto the **Angle** tile in the Marks card.



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

5. Enlarge the view: To make the pie chart legible, look at the top toolbar menu, find the fit dropdown menu (which says Standard by default), and switch it to **Entire View**.



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

Step C :- Designing Scatter Plots to Identify Relationships

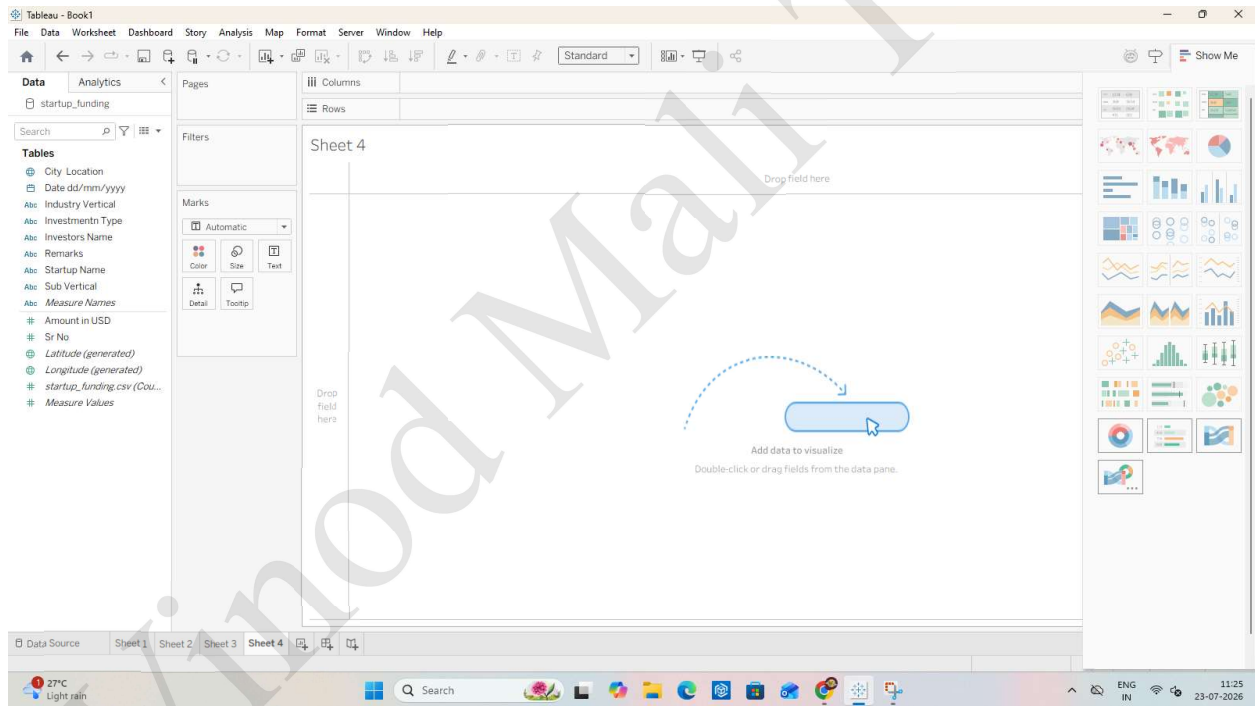
1. Cleaning the Data for Scatter Plots

A scatter plot maps relationships between two different numerical measures. However, in this dataset, we only have one explicit primary numeric field: Amount in USD.

To create a meaningful scatter plot, we will instruct Tableau to generate a second numerical measure dynamically by counting the number of times a startup name appears (representing the total number of funding rounds a startup went through).

2. Building the Scatter Plot

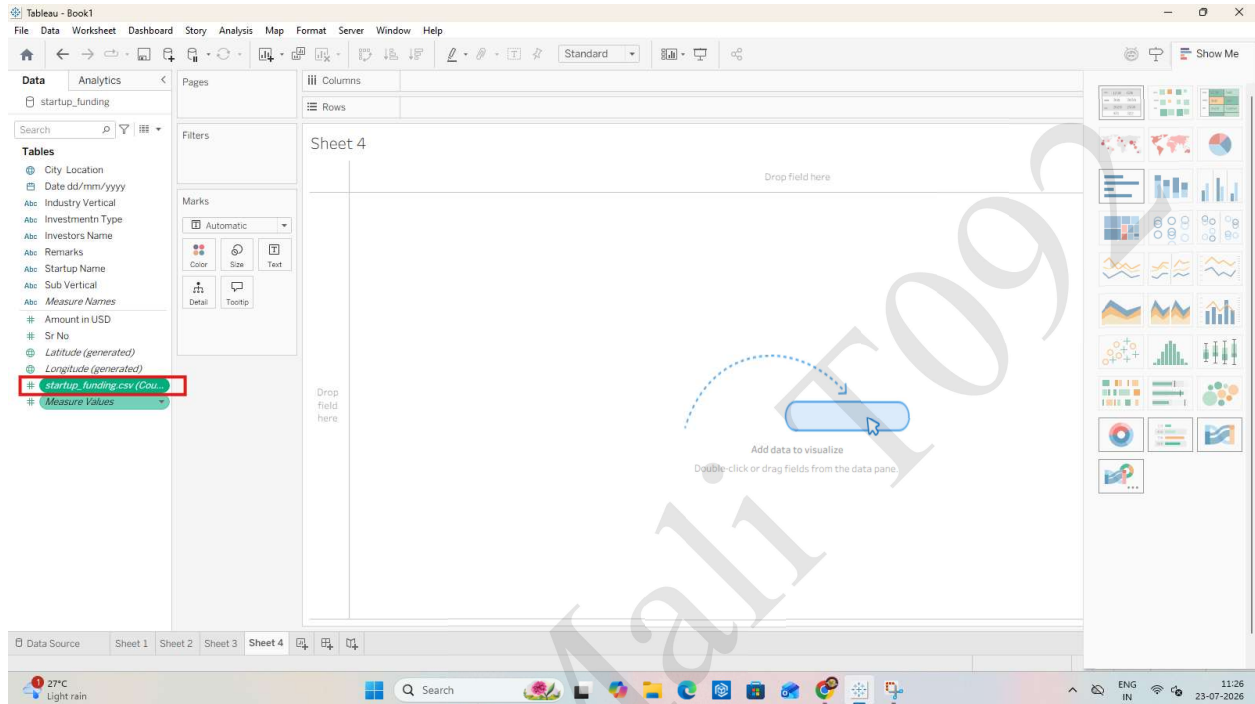
1. Click the **New Worksheet** icon at the bottom of your window to open a fresh canvas (**Sheet 4**).



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

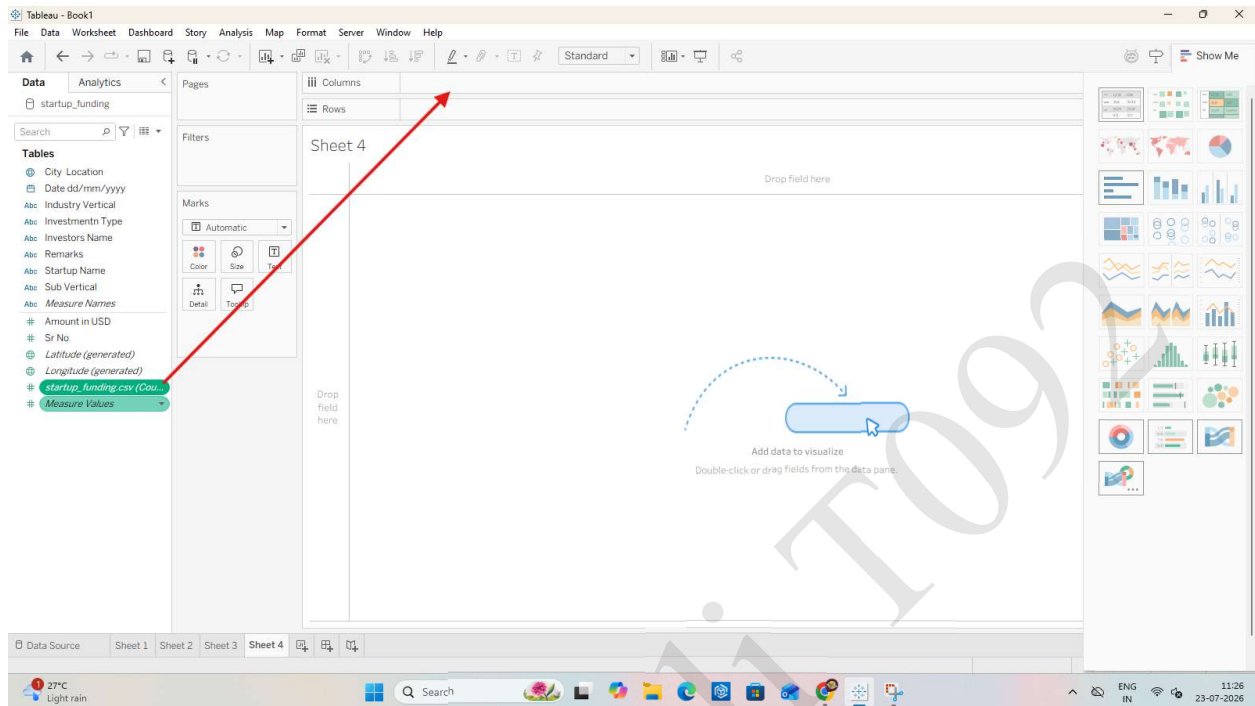
2. Look down to the bottom of your Data Pane list under the green measures section. Locate the automatic system measure field called **startup_funding.csv (Count)** (in older versions of Tableau, this is called **Number of Records**).



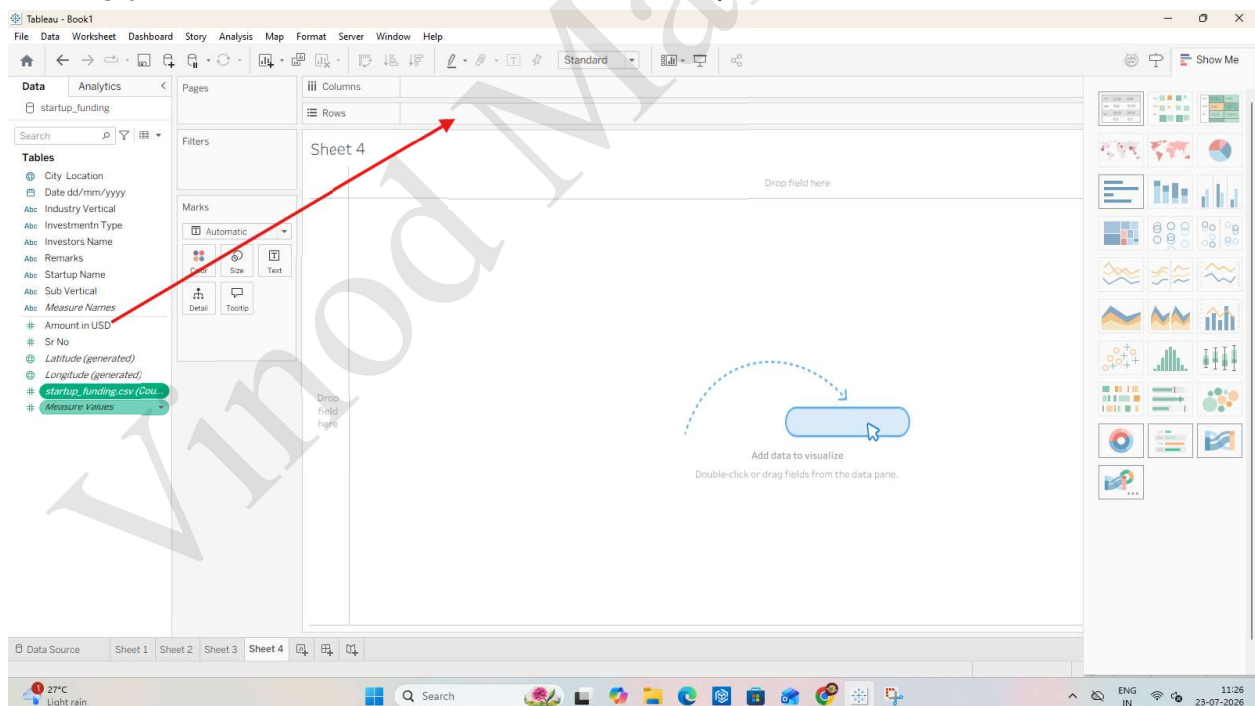
3. Drag that Count or Number of Records measure and drop it onto the Columns shelf.

Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

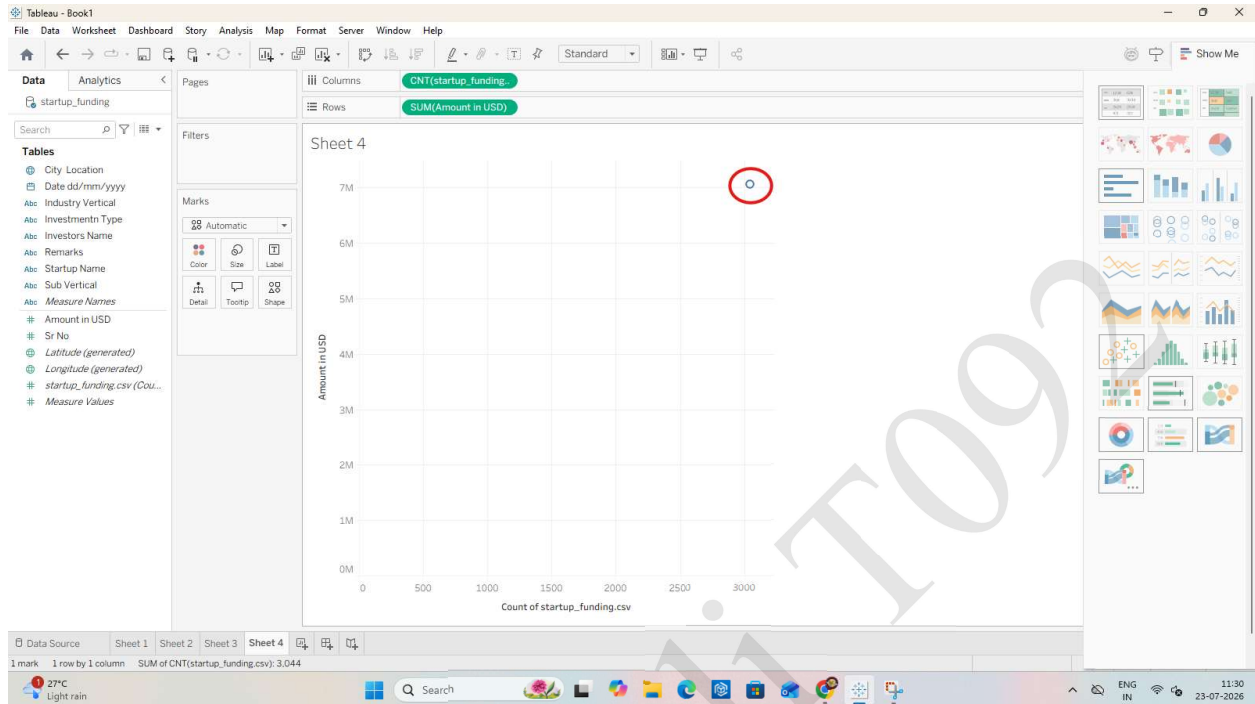


4. Drag your Amount in USD measure and drop it onto the Rows shelf.

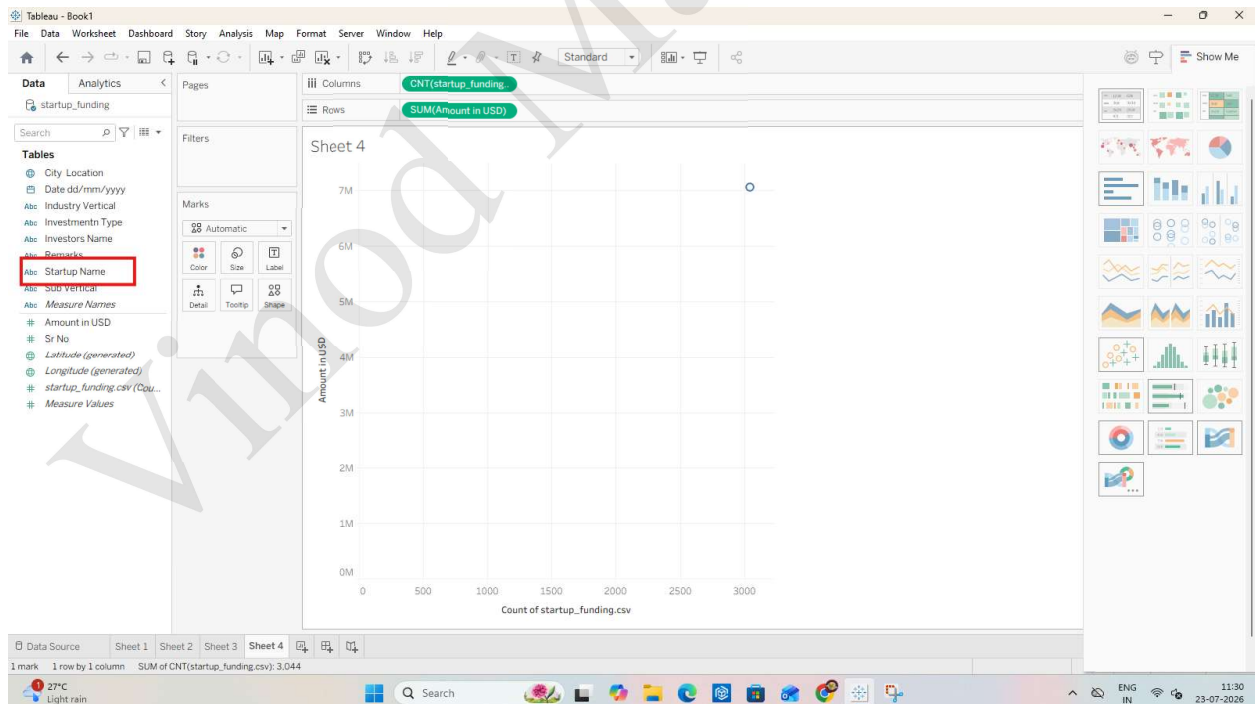


Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3



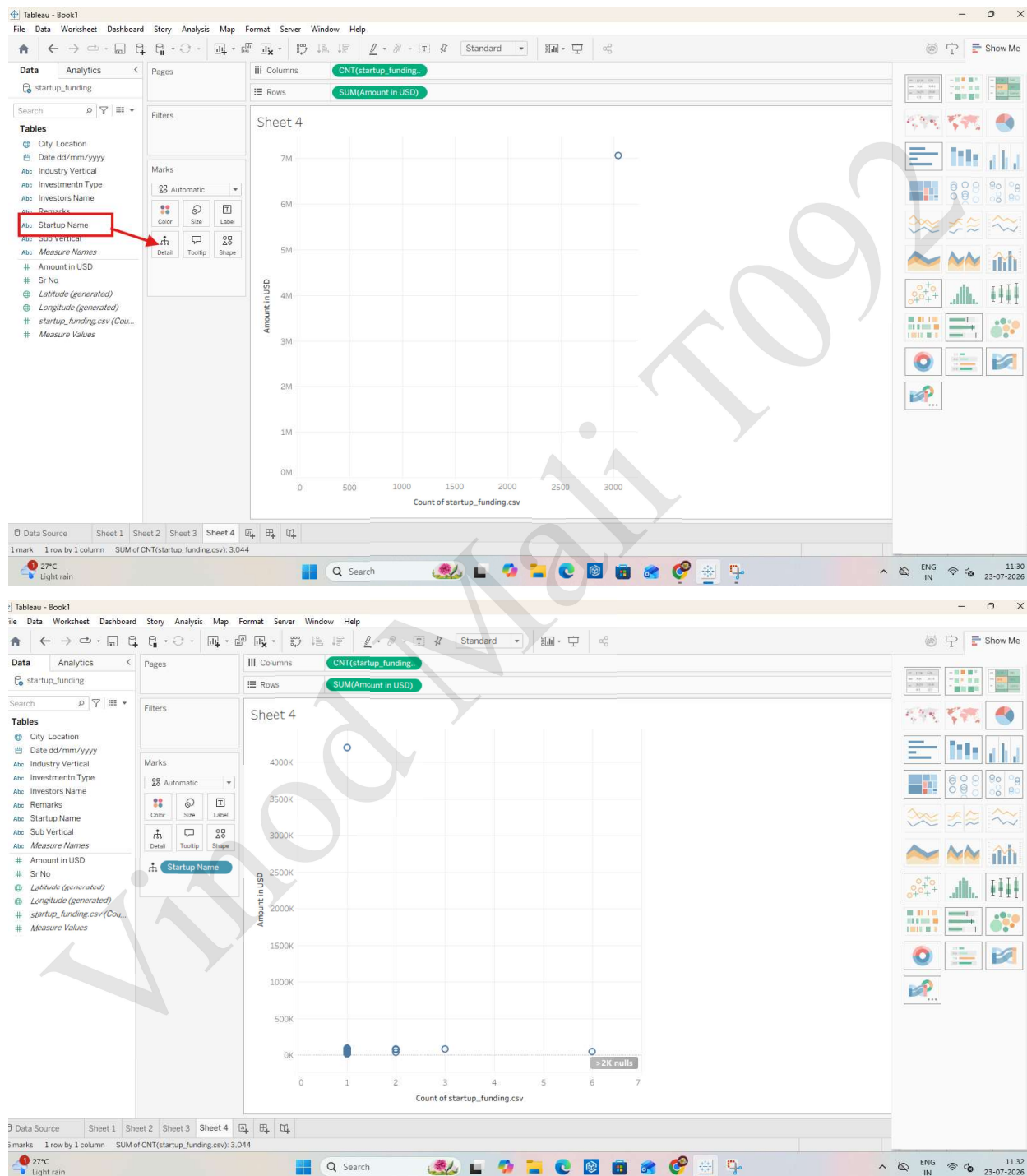
5. To break this single point down into individual startups, find the Startup Name dimension in the Data Pane.



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

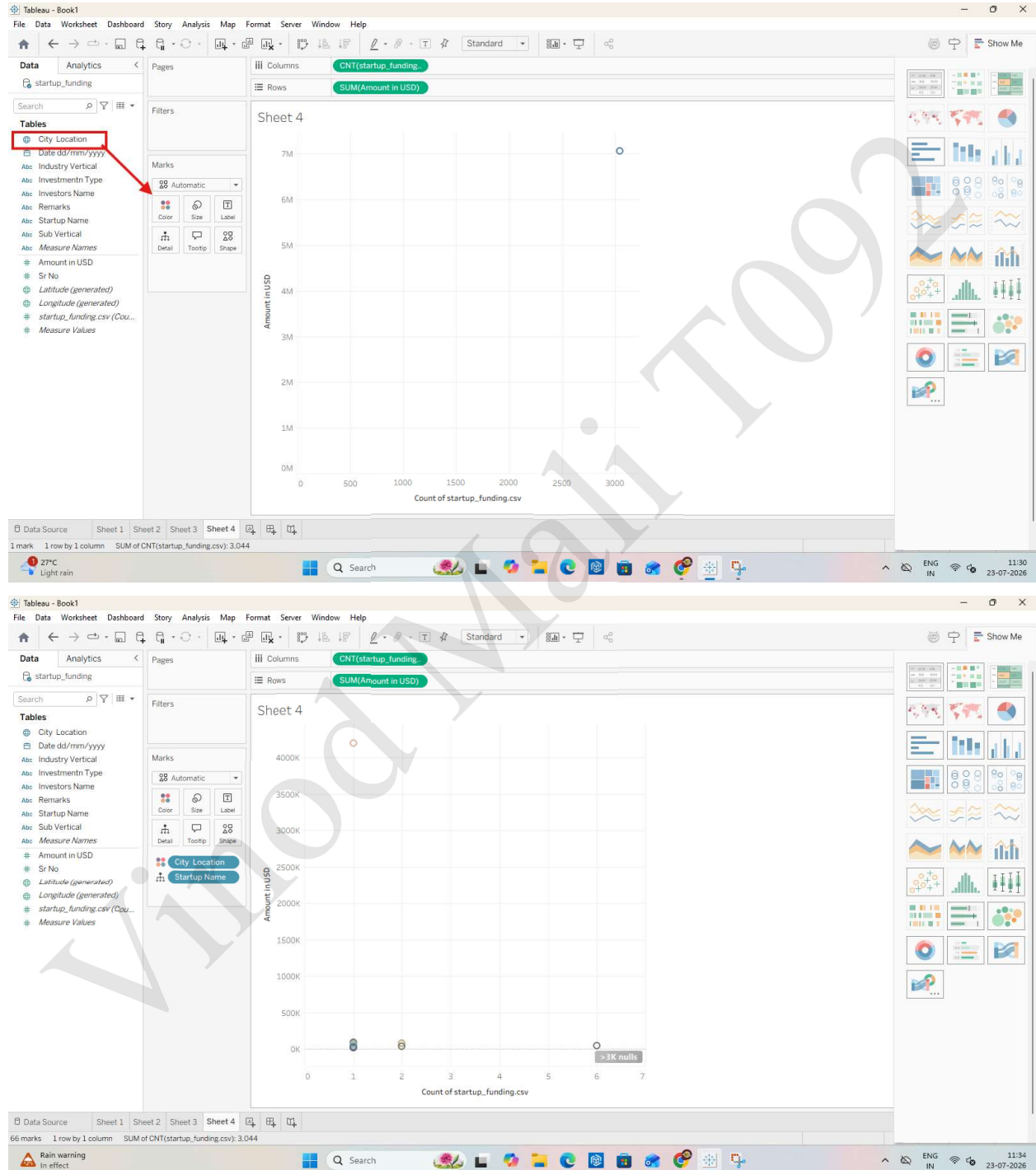
6. Drag Startup Name and drop it directly onto the Detail tile within the Marks card.



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

7. Adding Context with Color: Take the City Location dimension from your Data Pane and drop it onto the Color tile in the Marks card.



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

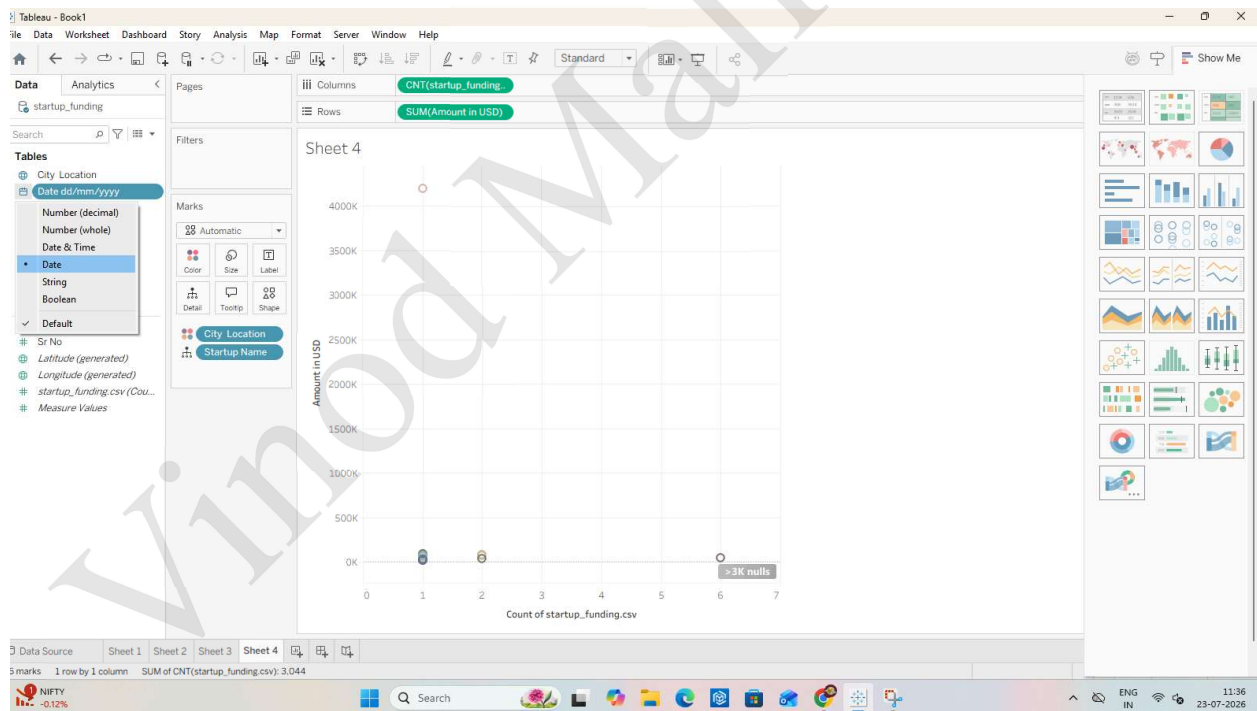
Step D: Creating Dual-Axis and Combination Charts

1. Preparing the Fields

To build a dual-axis combination chart, we need a time element on our horizontal axis (\$X\$-axis) and two different metrics tracking along the vertical axis (\$Y\$-axis).

We will use the **Date dd/mm/yyyy** column for time. Before we begin, let's make sure Tableau treats this text field as a proper date:

1. Look at your left-hand Data Pane.
2. Find the field **Date dd/mm/yyyy** and click the small text icon (Abc) immediately to the left of its name.
3. Select **Date** from the dropdown menu to convert it into a structured time dimension.

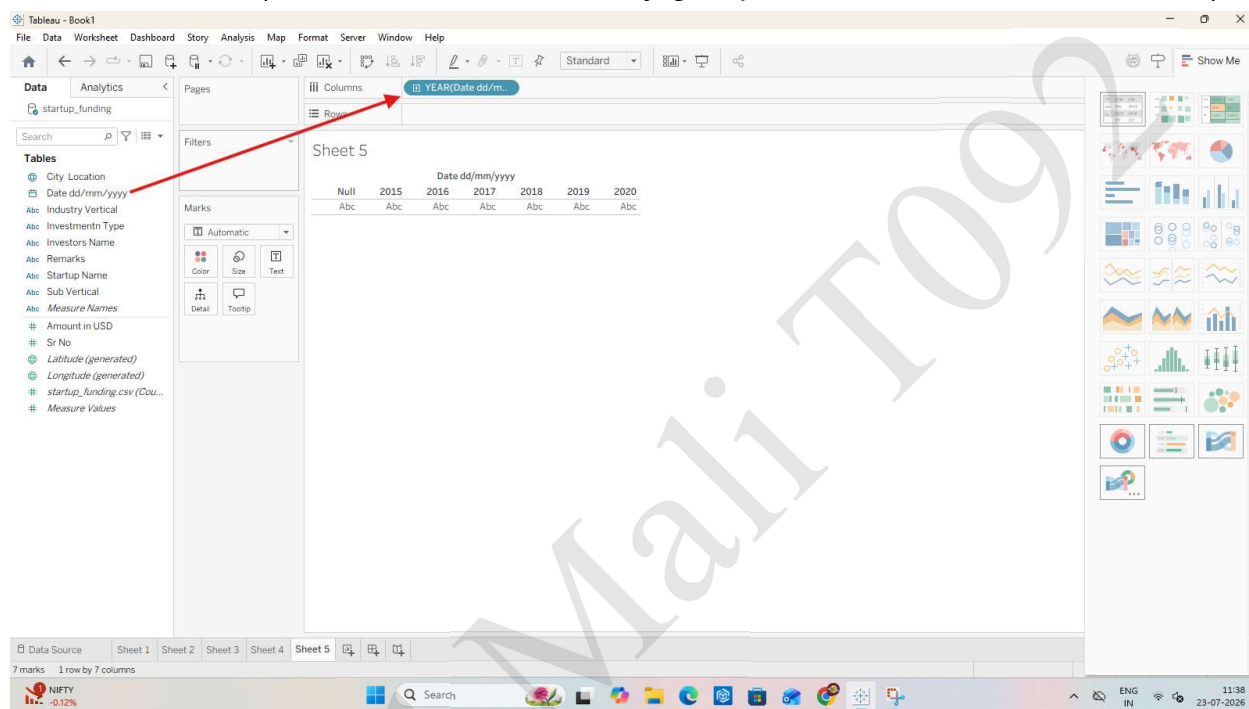


Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

2. Building the Dual-Axis Chart

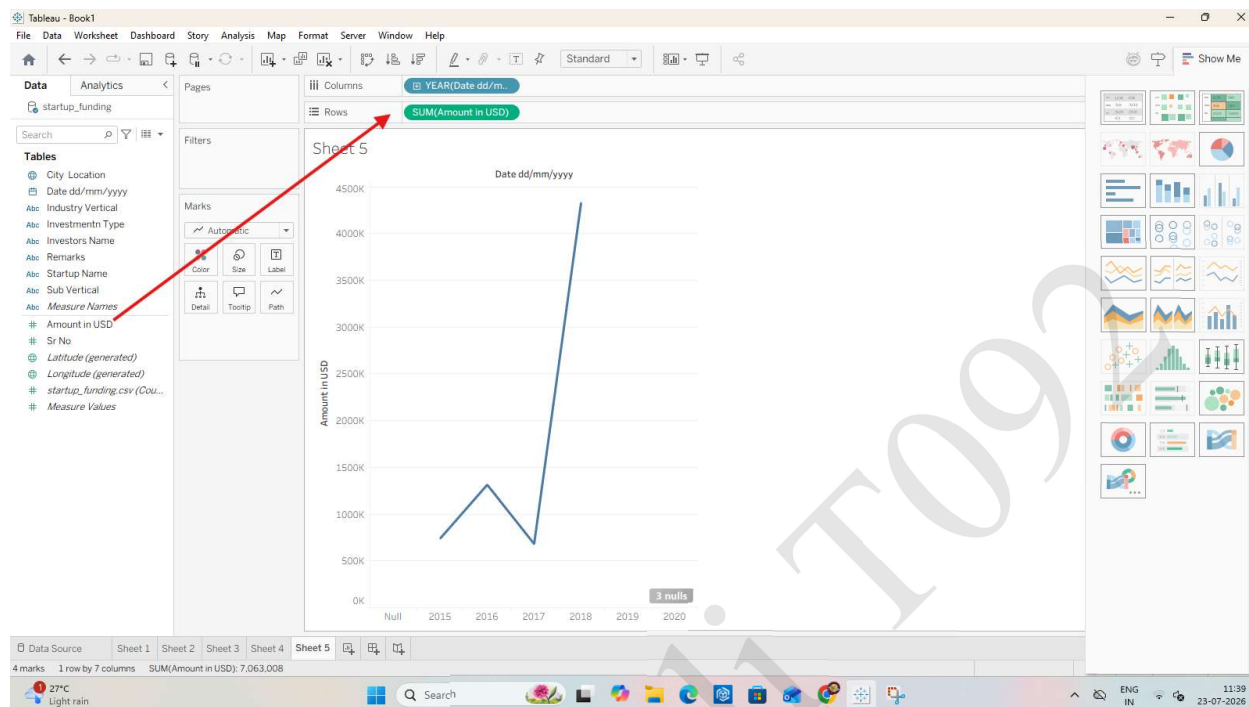
1. Click the **New Worksheet** icon at the bottom of the screen to open a fresh canvas (**Sheet 5**).
2. Drag your newly formatted Date dd/mm/yyyy dimension and drop it onto the **Columns** shelf. (Tableau will automatically group the dates into discrete **Years**).



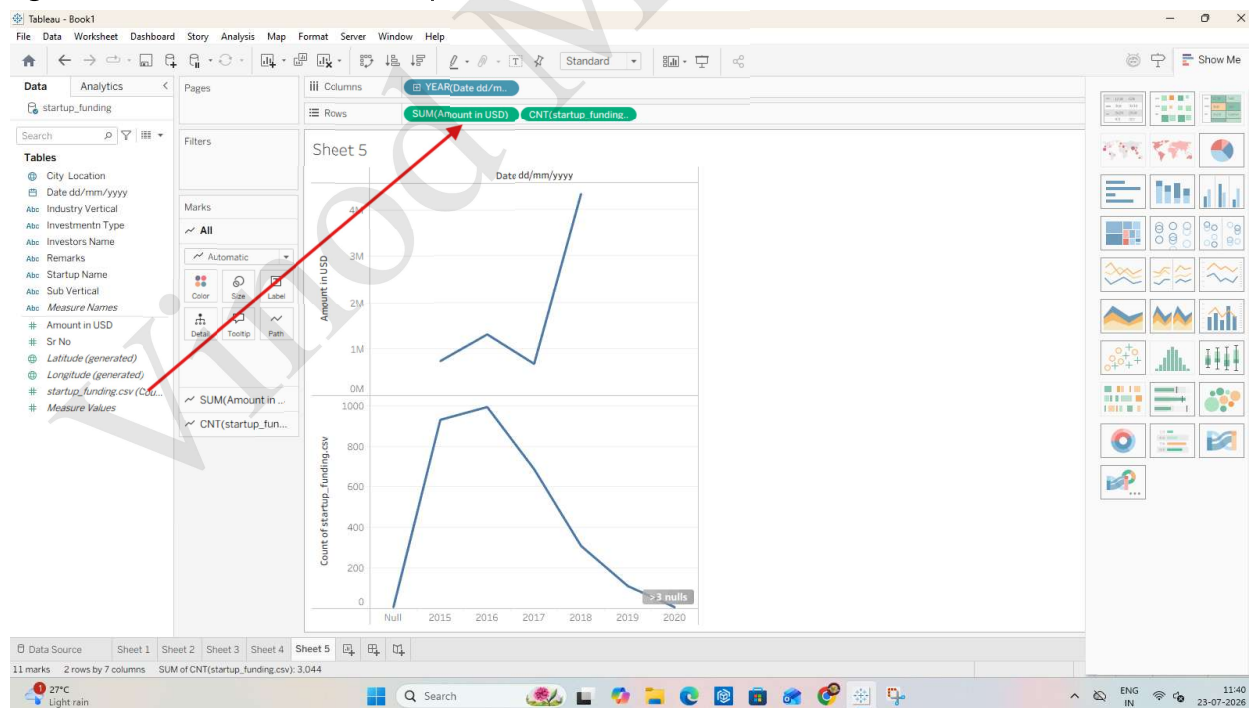
3. Drag the Amount in USD measure from the Data Pane and drop it onto the Rows shelf.

Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3



Now, take the automatic system measure startup_funding.csv (Count) (or Number of Records) and drag it onto the Rows shelf, placing it directly to the right of the Amount in USD pill.

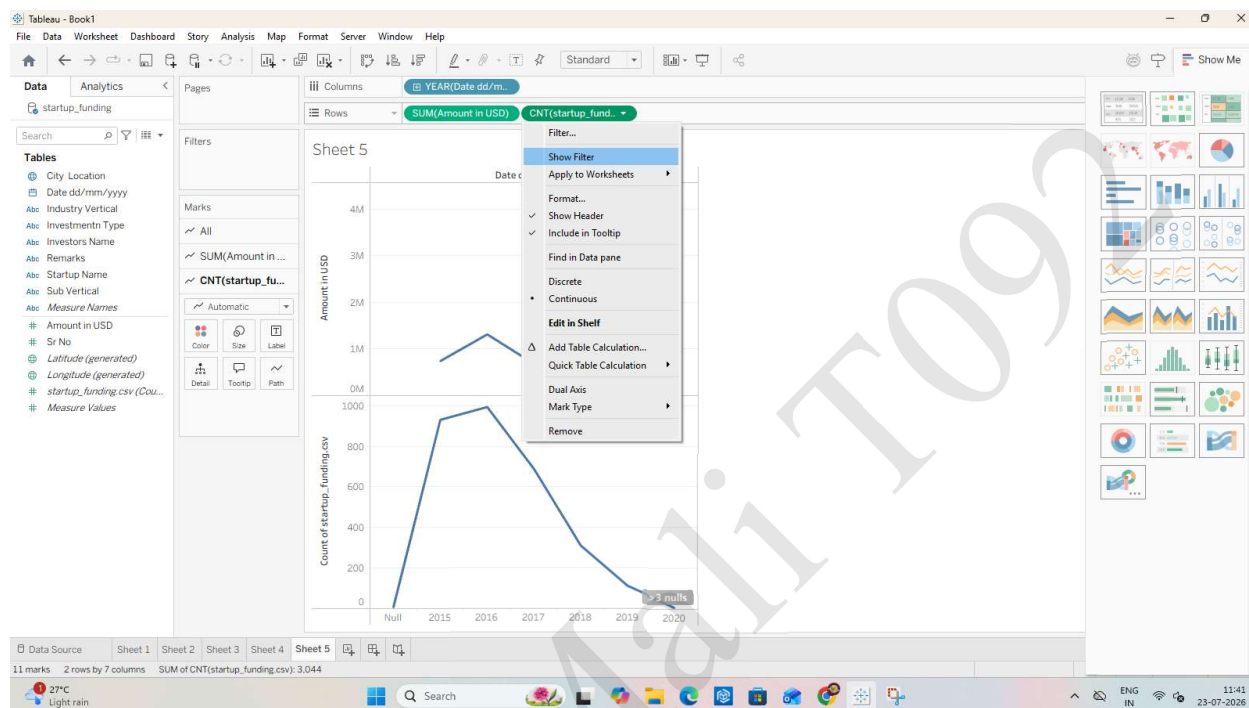


Name :- Vinod Manohar Mali
Roll no :- T092

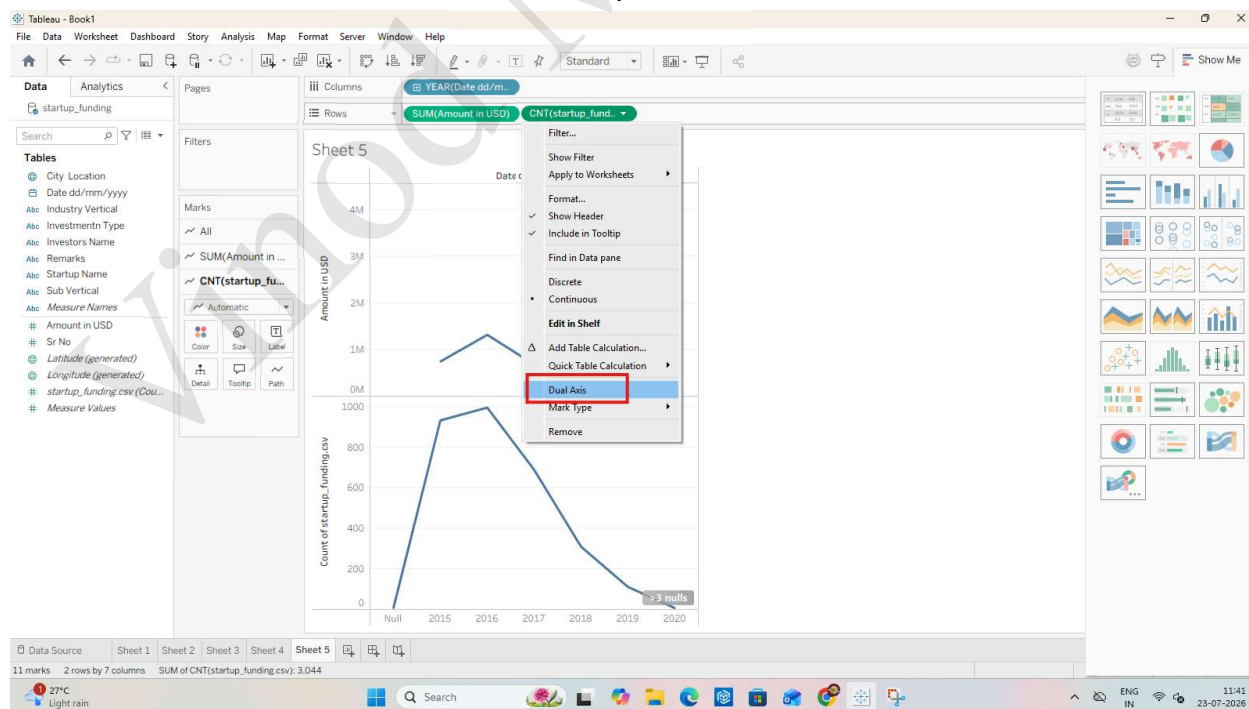
Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

3. Combining and Formatting the Visuals

1. Right-click the second measure pill (startup_funding.csv (Count)) sitting up on the Rows shelf.

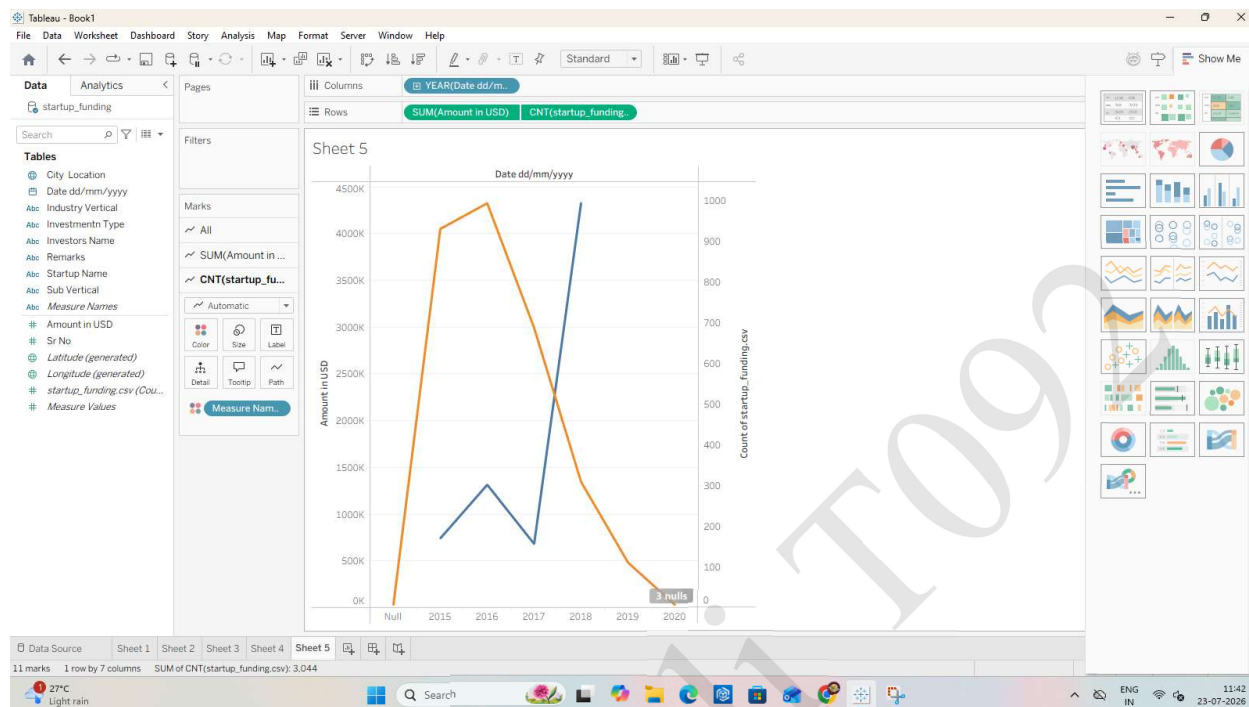


2. Select Dual Axis from the context dropdown menu.

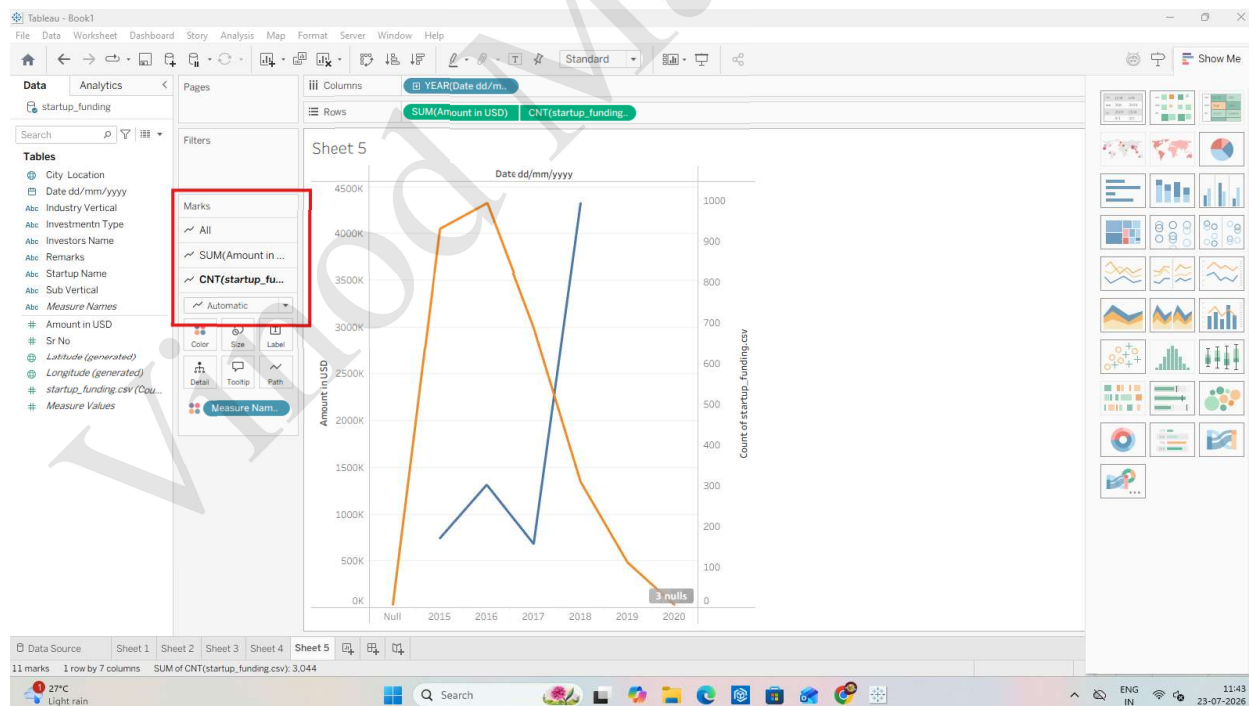


Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3



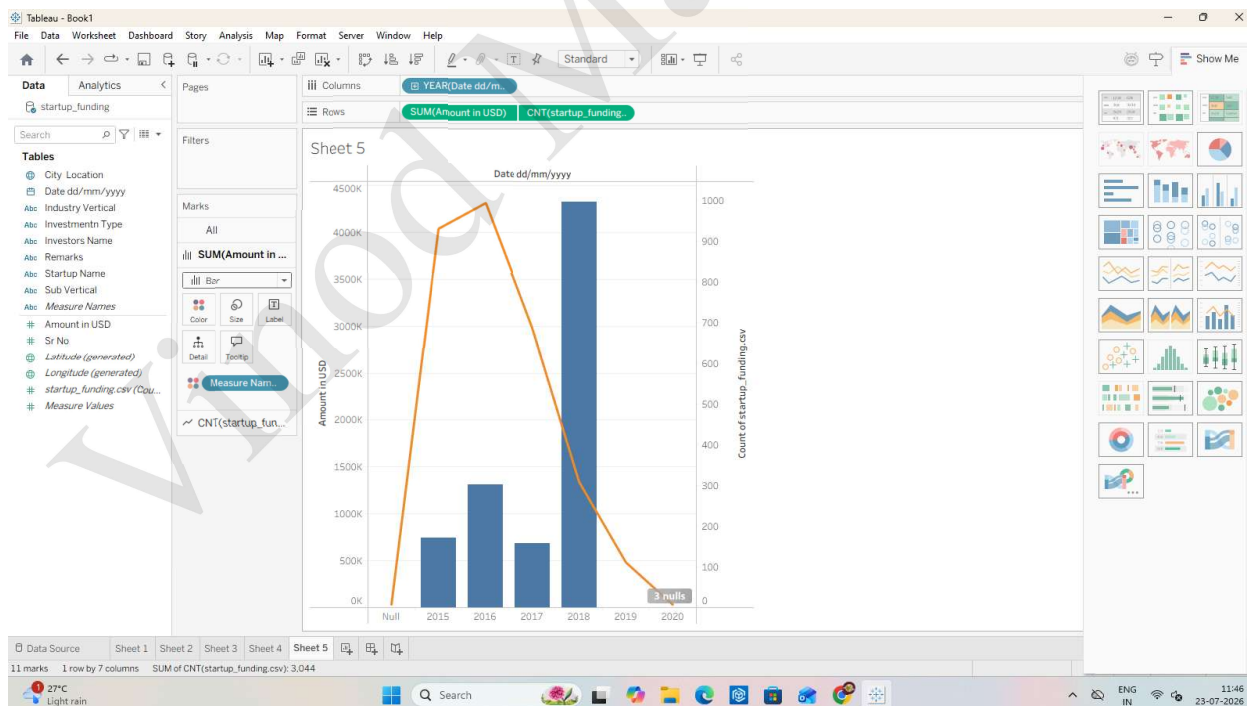
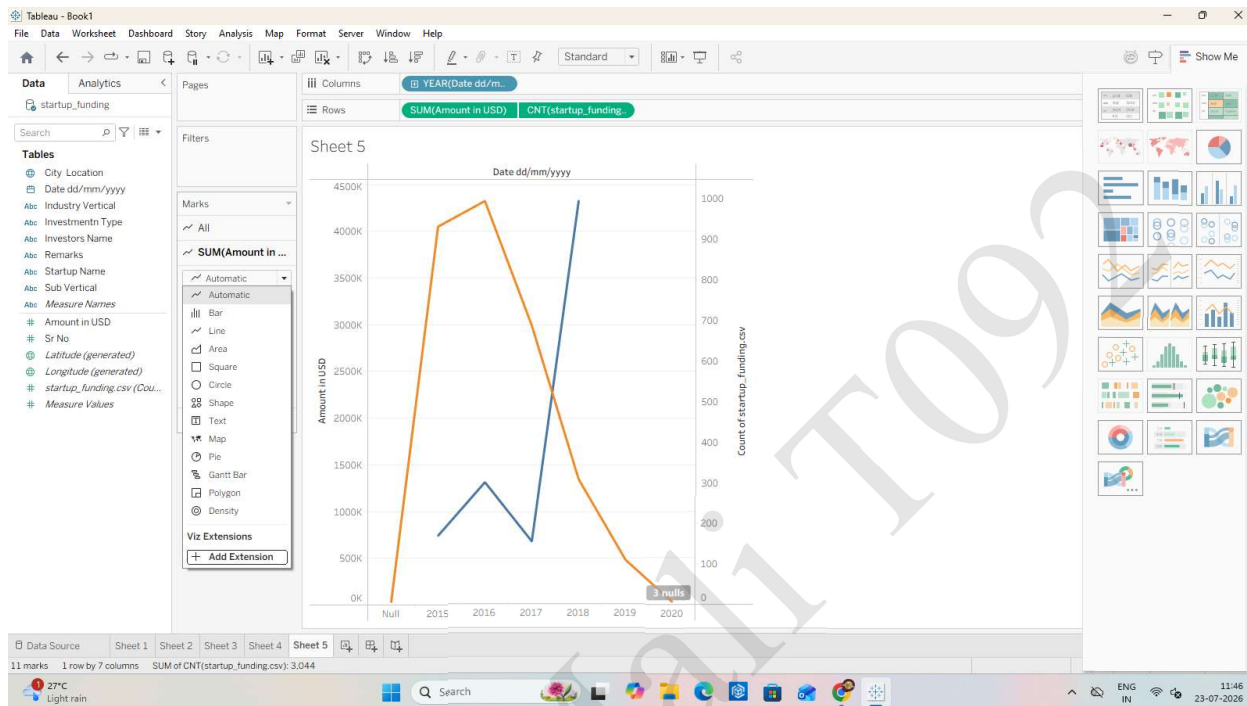
3. Look at your Marks Card area. Notice that it has split into three distinct tabs: All, SUM(Amount in USD), and CNT(startup_funding.csv).



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

4. Click directly on the SUM(Amount in USD) tab in the Marks card. Change its visual type dropdown selection from Automatic to Bar.



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

Step E: Developing Histograms for Distribution Analysis

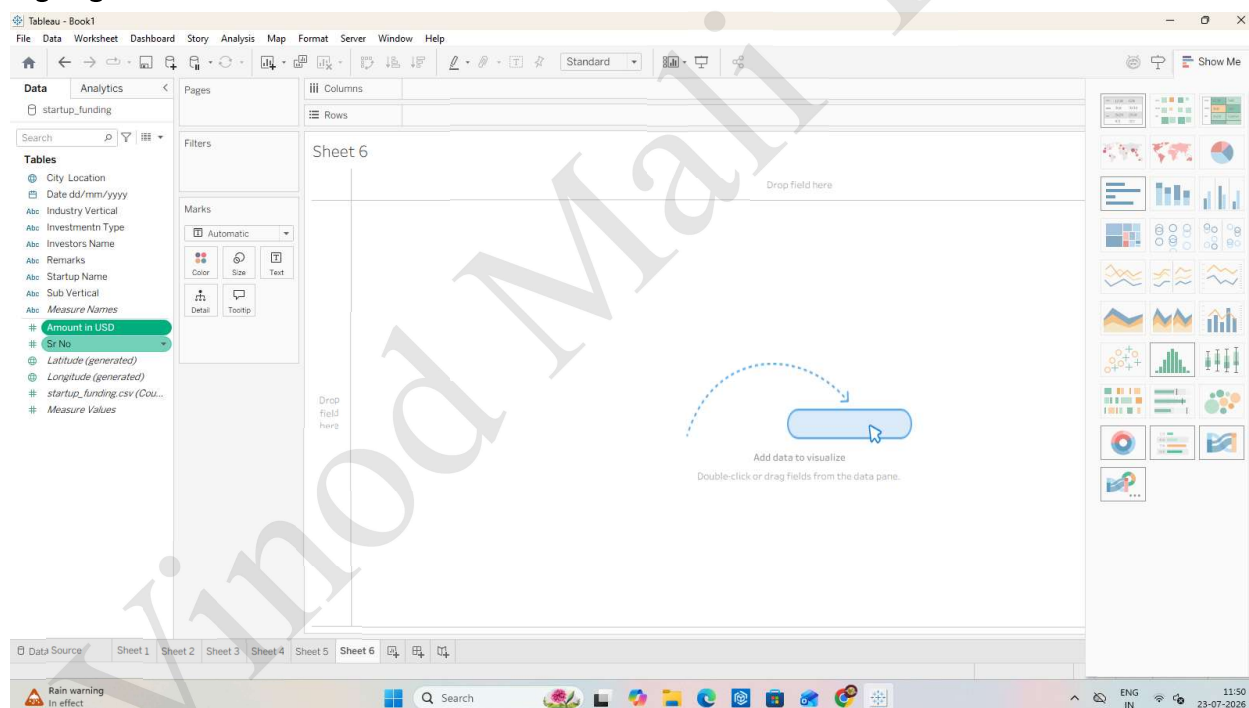
1. Understanding Histograms in Tableau

A histogram splits a single continuous numerical measure into equal, consecutive intervals called Bins (e.g., separating funding amounts into bins of 0–10M, 10M–20M, etc.) and displays the frequency count of records falling within each bucket as a bar chart.

Because a histogram relies on a single continuous field to build its own bins, we don't manually drop fields onto both shelves like a normal bar chart.

2. Building the Histogram

1. Click the New Worksheet icon at the bottom of the screen to open a fresh canvas (Sheet 6).
2. Go to the Data Pane and click once directly on the Amount in USD measure to highlight it.



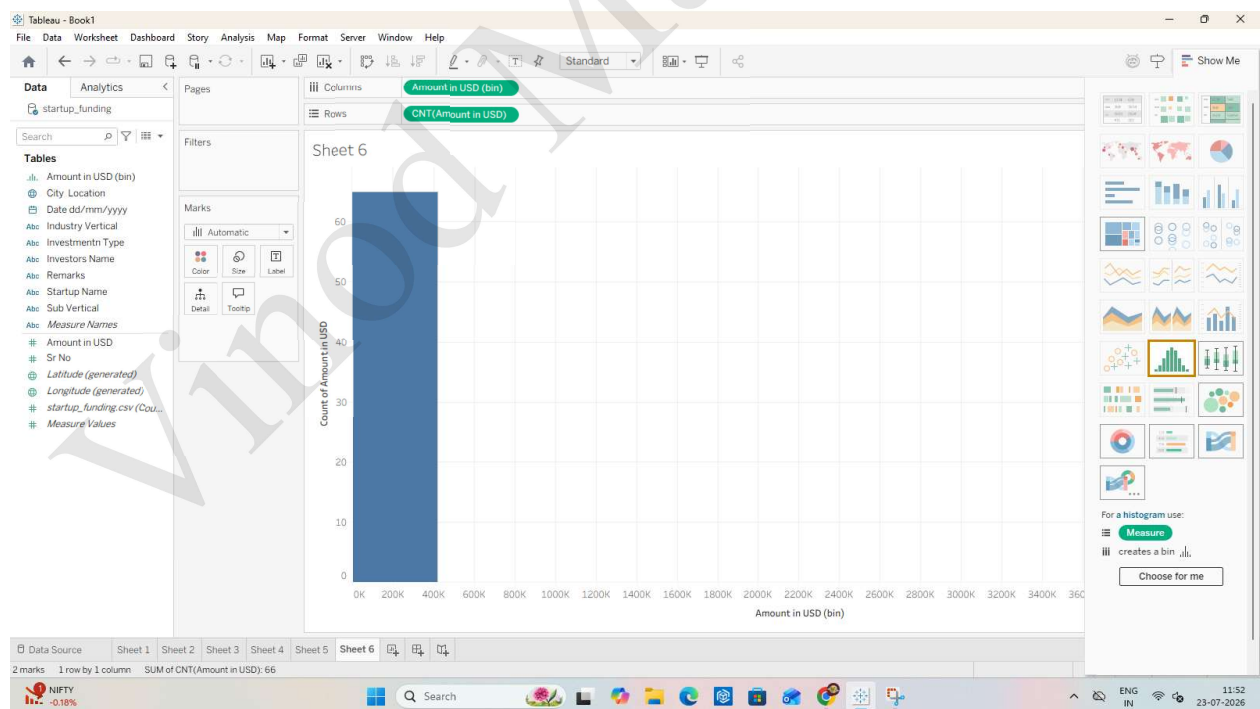
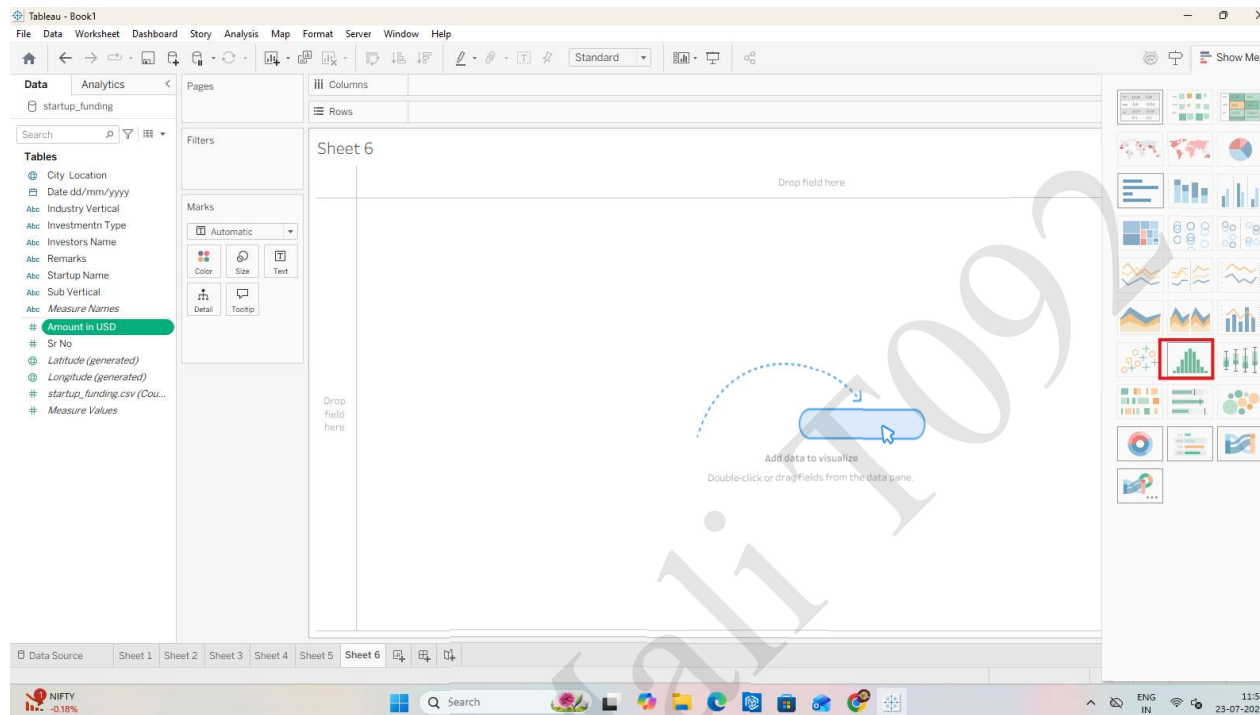
Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science

Subject :- Data Visualization with Power BI and Tableau

Practical no 3

3 . From the options, locate and click the Histogram icon (found in the middle row, displaying vertical green bars).

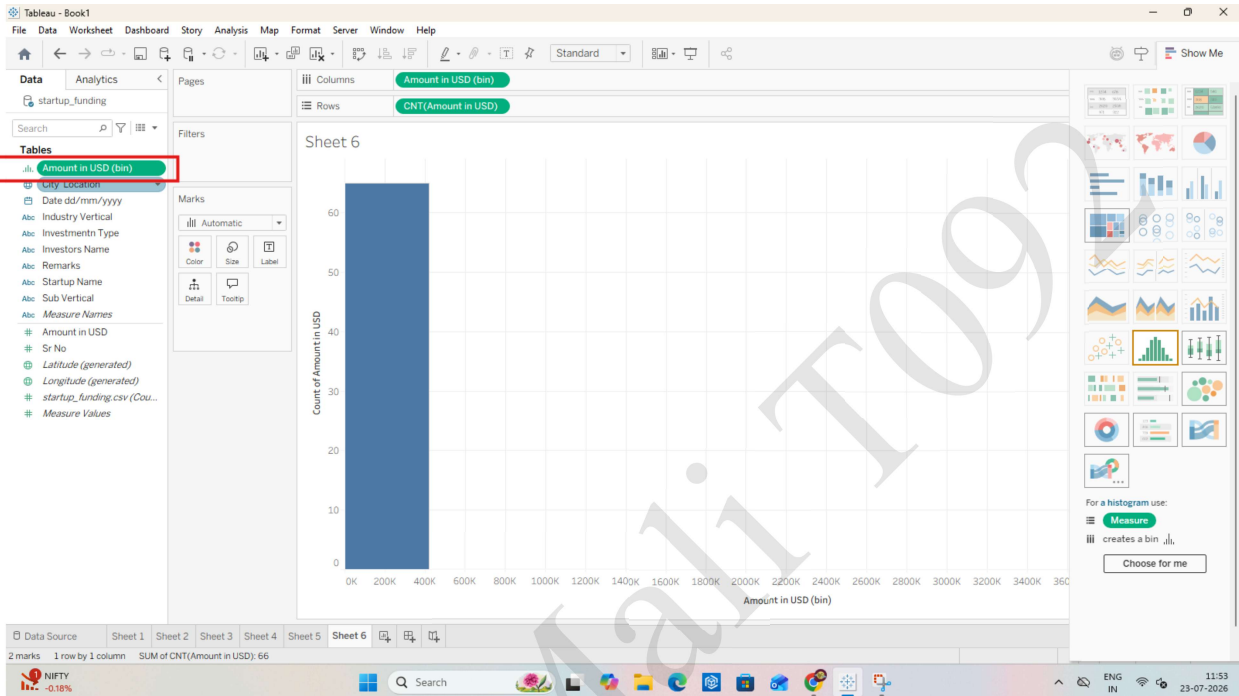


Name :- Vinod Manohar Mali

Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

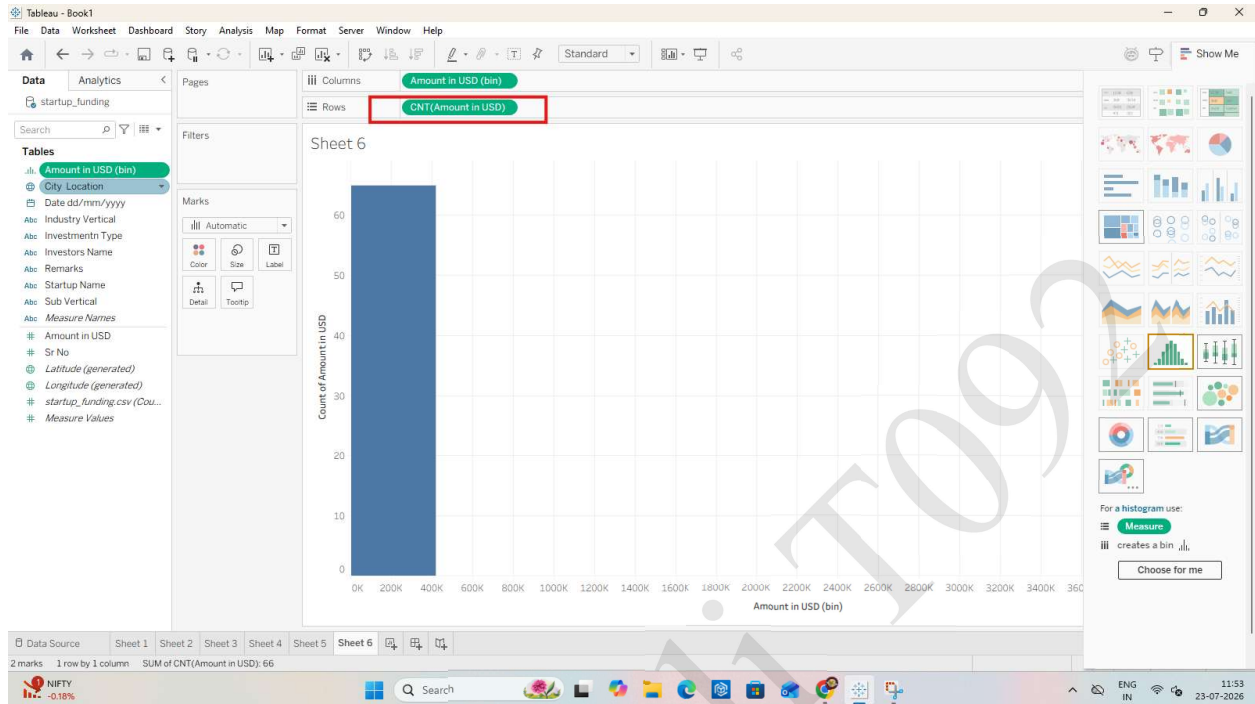
4. Look back at your Data Pane. Notice that Tableau has automatically generated a brand-new field under your dimensions list called Amount in USD (bin) and placed it onto your Columns shelf.



6. Look at the Rows shelf. Tableau has automatically placed the aggregated measure CNT(Amount in USD) there to track record counts.

Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3



Step F: Building Box-and-Whisker Plots

1. Cleaning the View for Box Plots

A Box-and-Whisker plot is a statistical chart used to show the distribution of data values based on a five-number summary (Minimum, First Quartile, Median, Third Quartile, and Maximum).

Because box plots measure structural distribution across data points, we must tell Tableau to look at the individual funding records rather than squishing everything into a single consolidated sum bar.

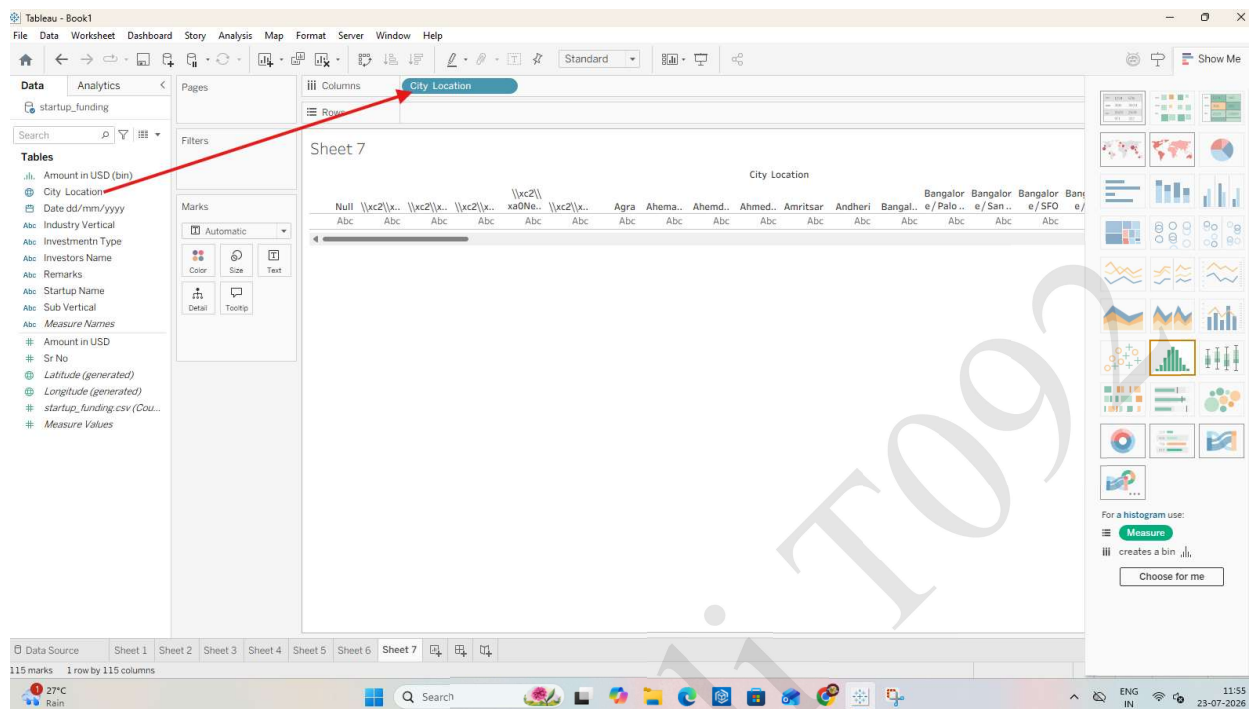
2. Building the Box-and-Whisker Plot

1. Click the New Worksheet icon at the bottom of the screen to open a fresh canvas (**Sheet 7**).
2. Drag the City Location dimension from the Data Pane and drop it onto the **Columns** shelf.

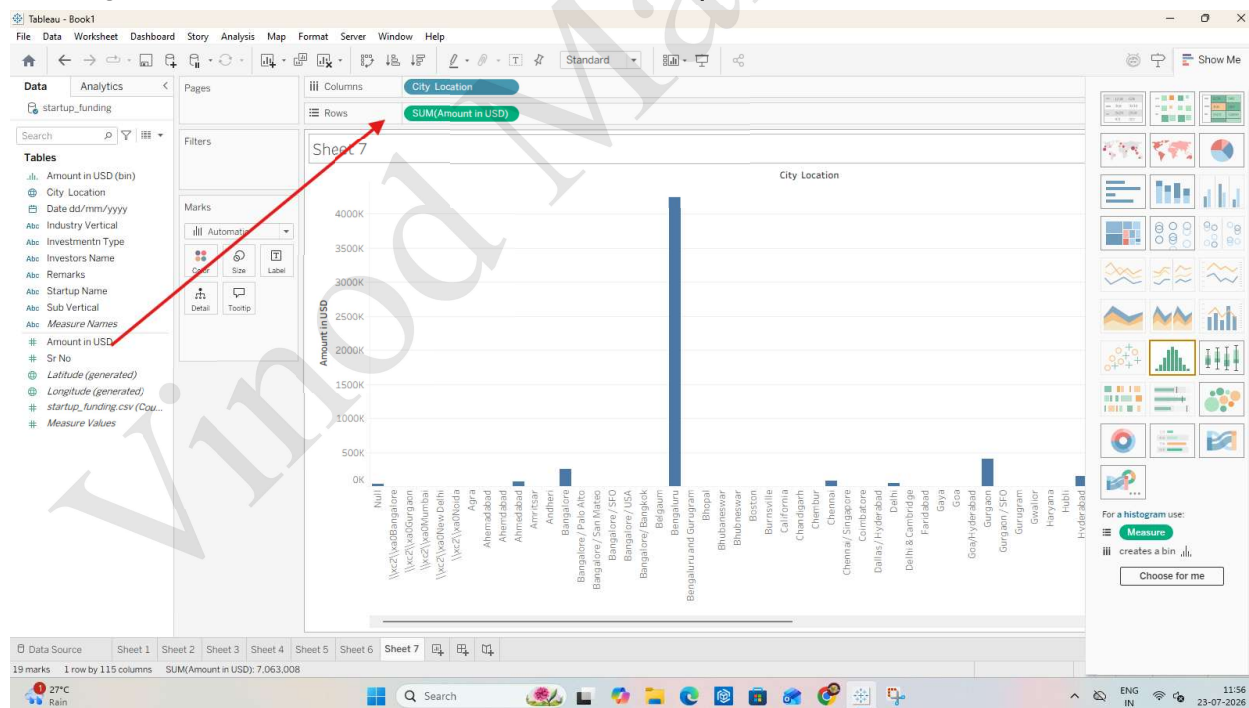
Name :- Vinod Manohar Mali

Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3



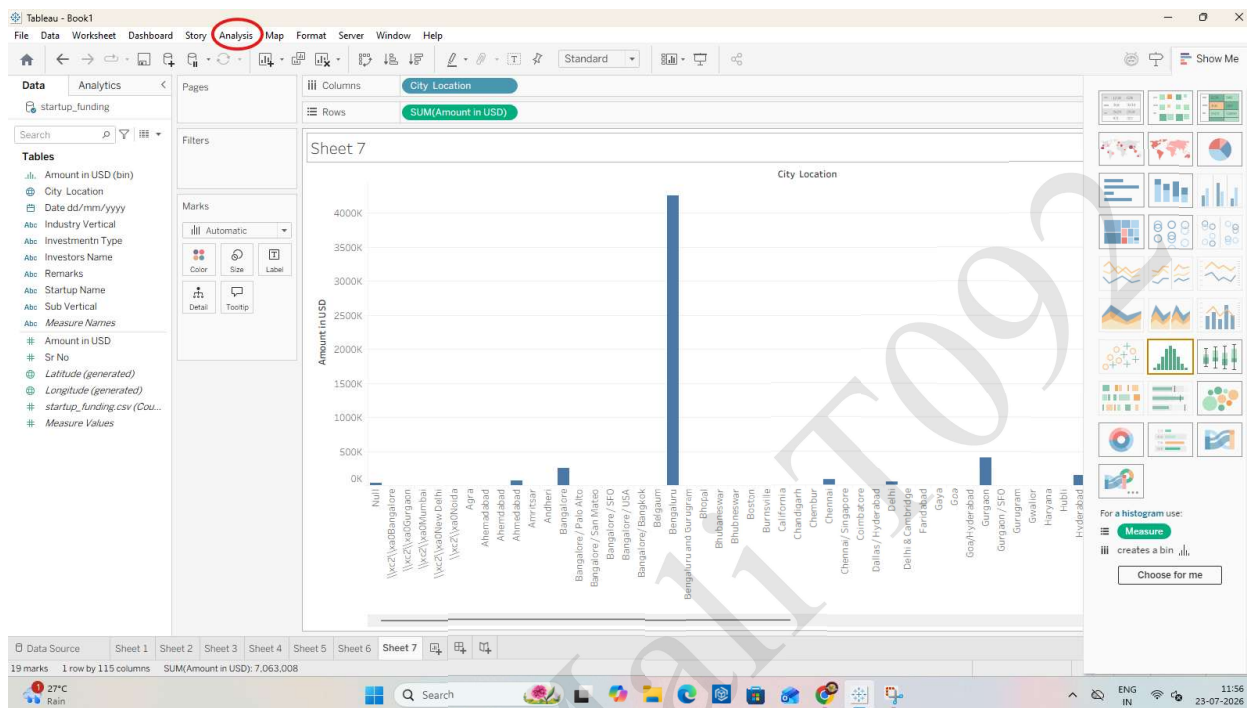
3. Drag the Amount in USD measure and drop it onto the Rows shelf.



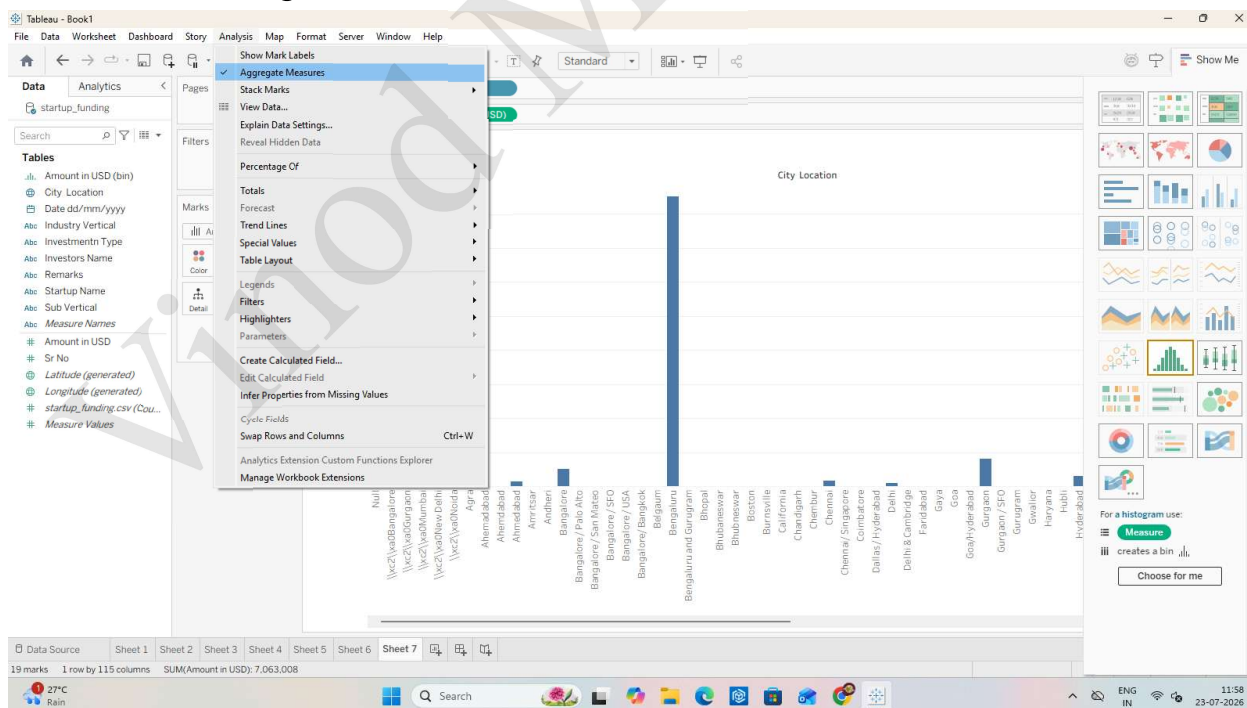
Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

4. Disaggregate the Data Points: Go to the top menu bar click Analysis, and uncheck Aggregate Measures.



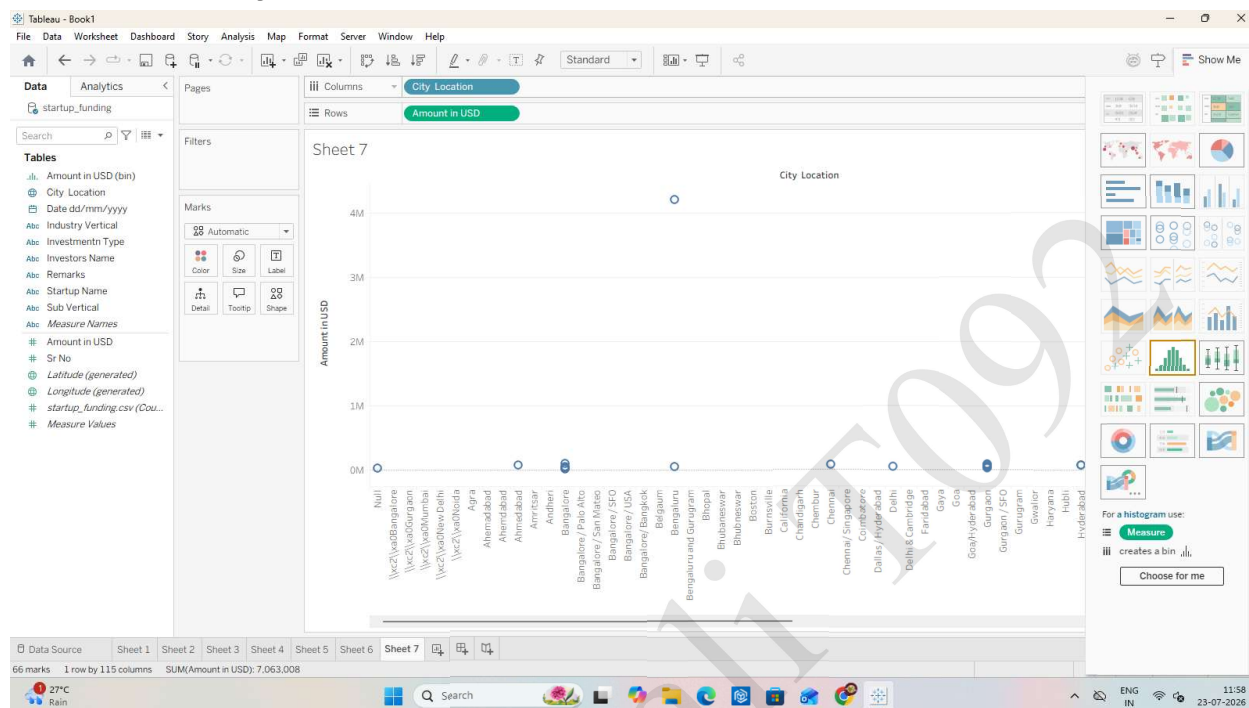
Before Unchecking



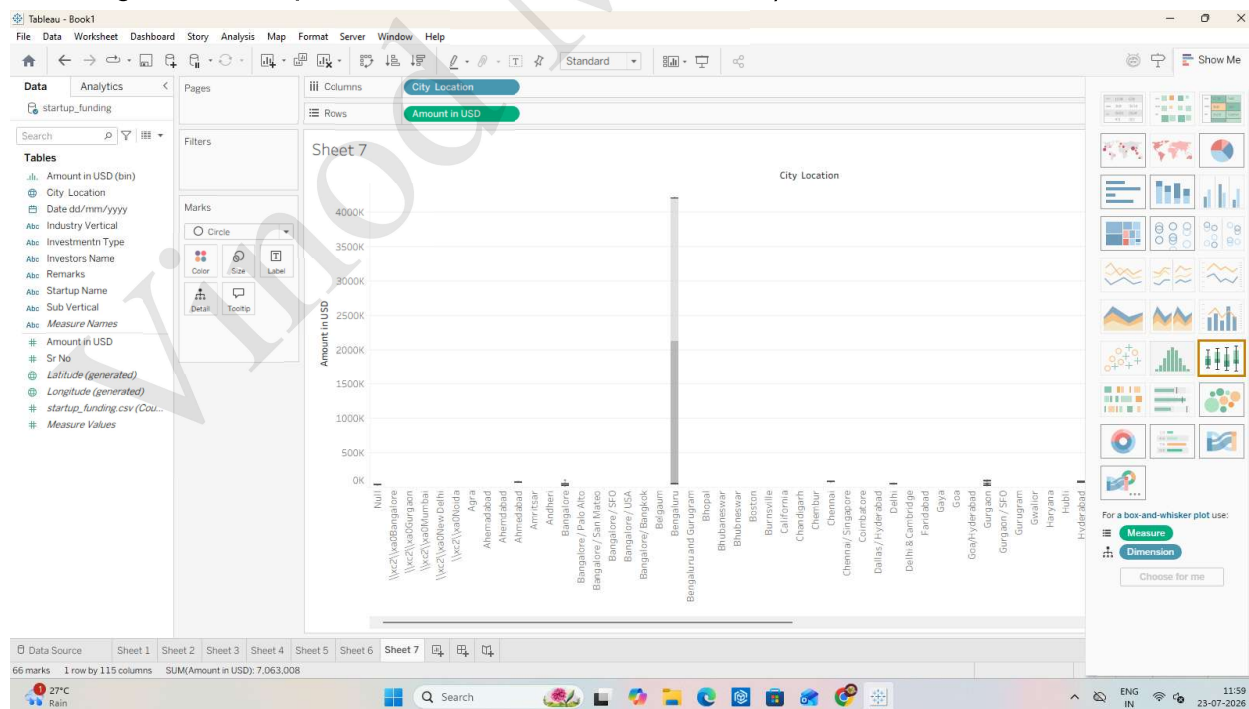
Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

After Unchecking



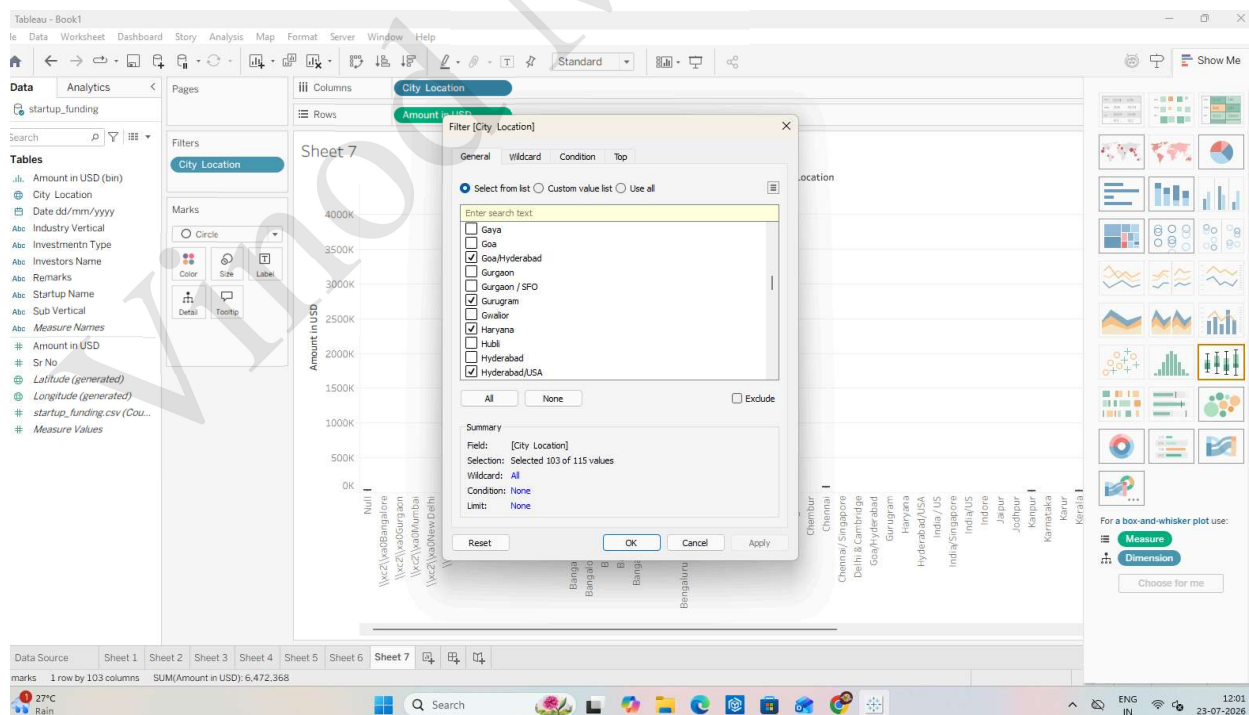
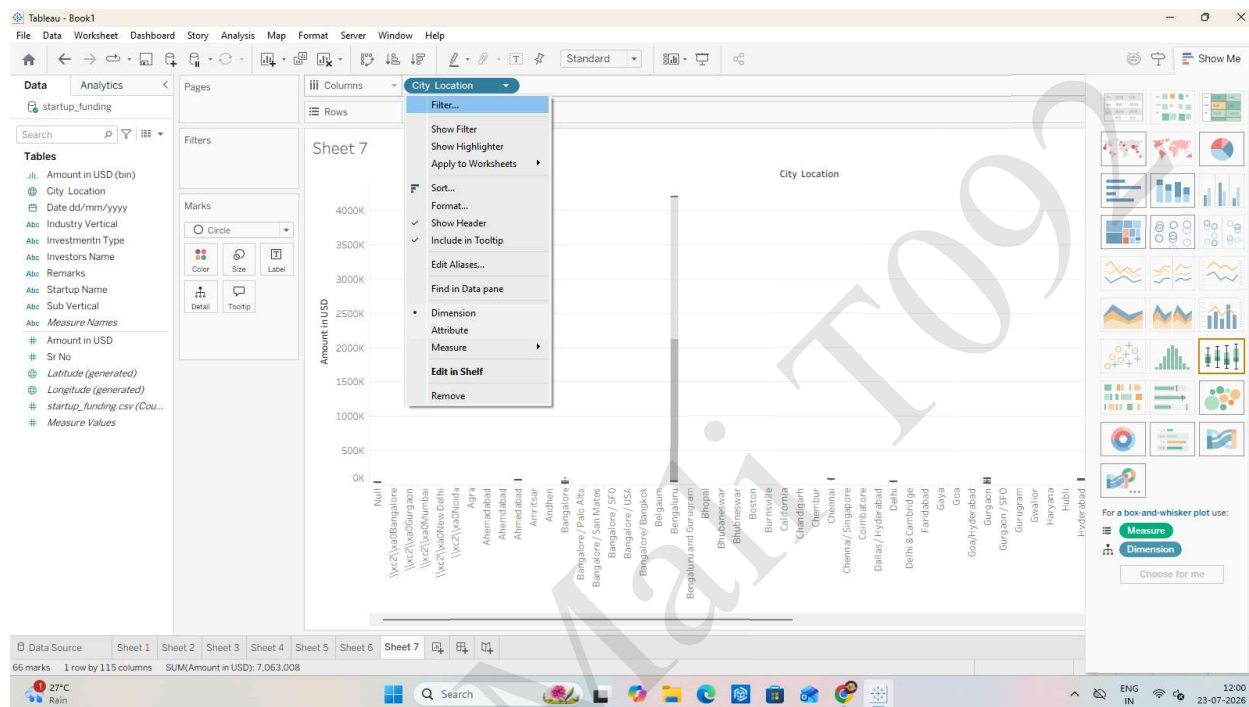
5. Add the Box Plot: Look at the top-right corner of the window, click the Show Me button, and select the Box-and-Whisker Plots icon (found in the middle row, showing a box shape with extended antenna lines).



Name :- Vinod Manohar Mali
Roll no :- T092

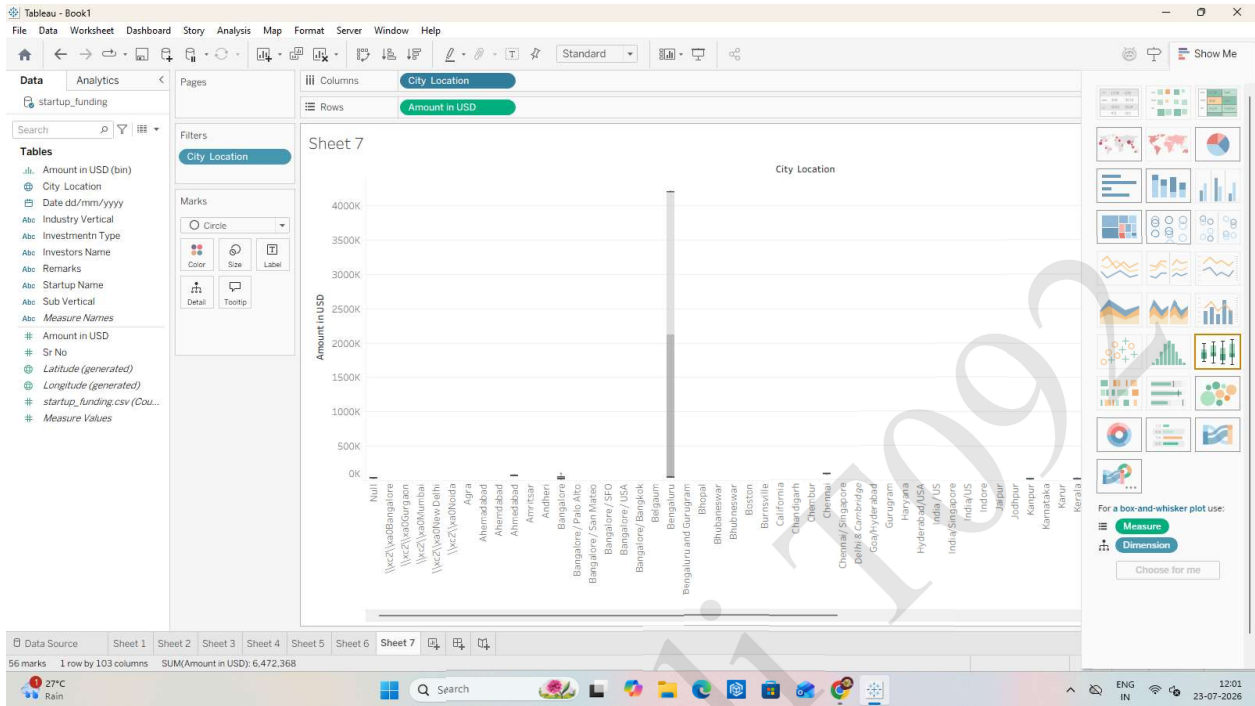
Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

6. Refining the Focus: Since there are many small towns with only one funding record, right-click the City Location pill on your Columns shelf, select Filter, choose only the top tier hubs (e.g., Bengaluru, Mumbai, New Delhi, Gurgaon), and click OK to clean up your distribution view.



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3



Step G: Using Gantt Charts and Advanced Date Visualizations

1. Understanding Gantt Charts in Tableau

A Gantt Chart is traditionally used to show project schedules and the duration of activities over time. In a standard Gantt chart, the horizontal axis (X-axis) represents a continuous timeline, marks are displayed as horizontal bars, and the length of each bar represents the duration of that specific event.

To build a meaningful duration layout using our funding dataset, we will calculate the time spread across funding dates or analyze milestone timelines. However, Tableau also allows us to use the Gantt Bar mark type to build advanced tracking timelines. Let's create an advanced chronological timeline of funding events.

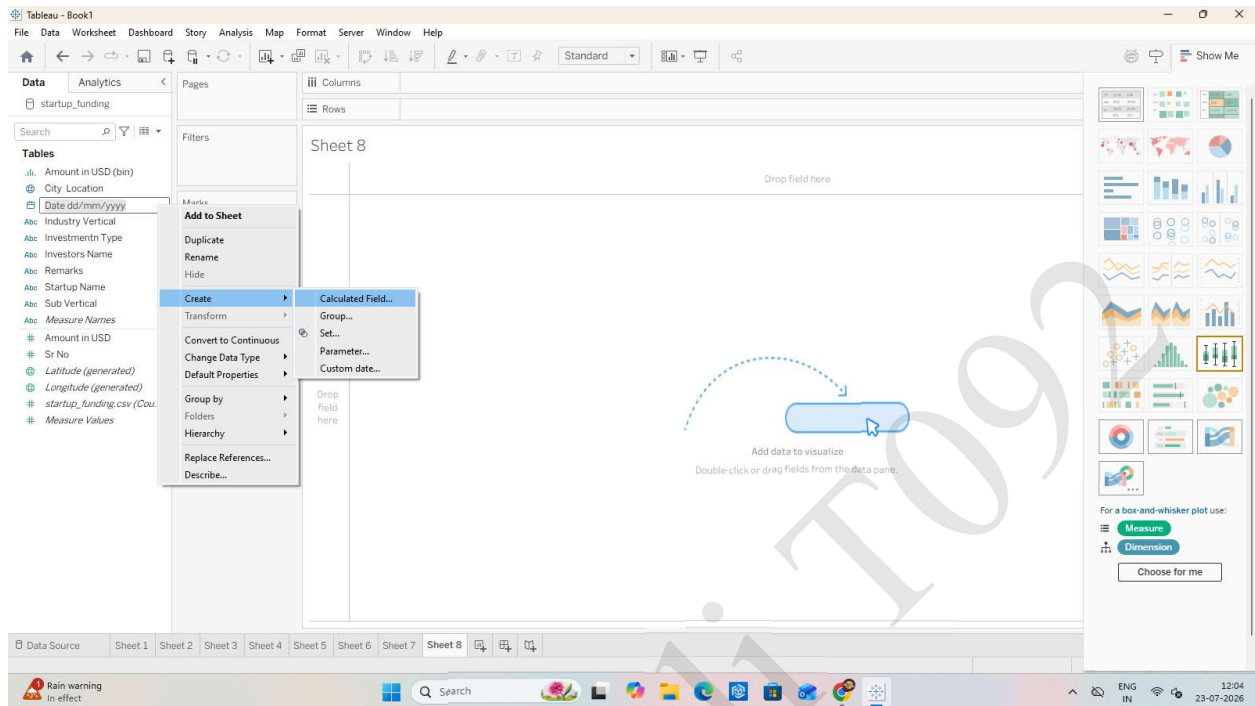
2. Building an Advanced Date Gantt Timeline

1. Click the New Worksheet icon at the bottom of the screen to open a fresh canvas (**Sheet 8**).
2. Look at your left Data Pane. Right-click on your Date dd/mm/yyyy field and select Create > Calculated Field.

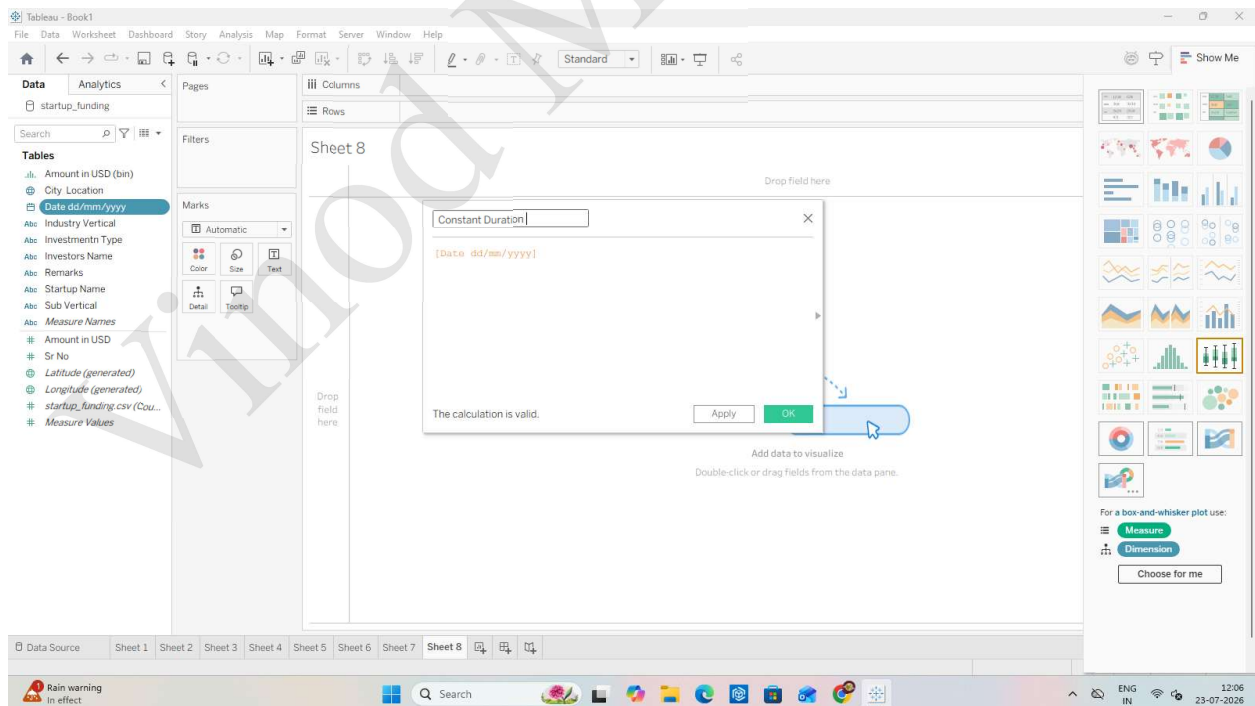
Name :- Vinod Manohar Mali

Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3



3. Name this new calculation Constant Duration. In the calculation box, type simply: 1 (This acts as a uniform placeholder length for our timeline blocks). Click OK.



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

4. Drag your continuous Date dd/mm/yyyy field and drop it onto the Columns shelf.

The first screenshot shows the Tableau interface with the 'Columns' shelf containing the field 'YEAR(Date dd/mm/yyyy)'. The 'Marks' shelf is set to 'Automatic'. The view displays a table with columns for the year (2015-2020) and rows for 'Null' and 'Abc'. The second screenshot shows the context menu for the 'YEAR(Date dd/mm/yyyy)' field on the Columns shelf. The 'Exact Date' option is selected, which changes the view to a date-based visualization. The table now shows dates (e.g., 2015, Q2 2015, May 2015) instead of just years.

Tableau - Book1

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Columns: YEAR(Date dd/mm/yyyy)

Rows:

Sheet 8

Date dd/mm/yyyy	2015	2016	2017	2018	2019	2020
Null	Abc	Abc	Abc	Abc	Abc	Abc
Abc	Abc	Abc	Abc	Abc	Abc	Abc

7 marks 1 row by 7 columns

Tableau - Book1

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Columns: YEAR(Date dd/mm/yyyy)

Rows:

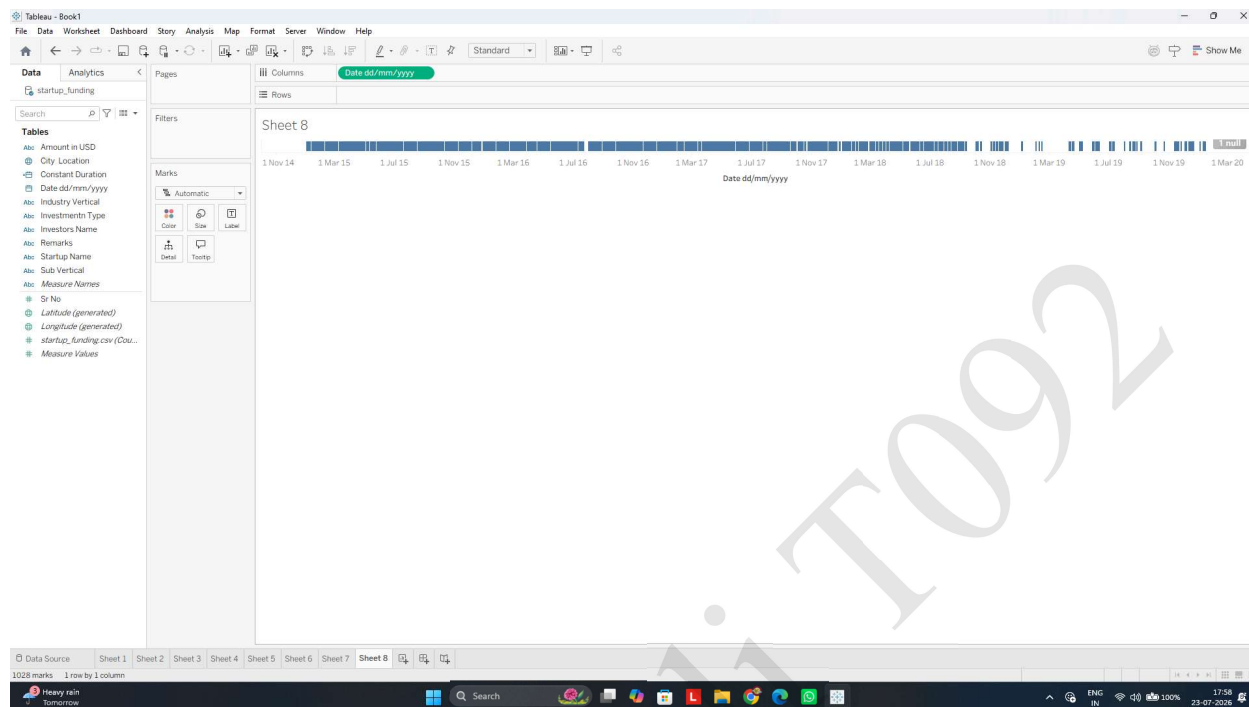
Sheet 8

Date dd/mm/yyyy	2015	2016	2017	2018	2019	2020
Null	Abc	Abc	Abc	Abc	Abc	Abc
Abc	Abc	Abc	Abc	Abc	Abc	Abc

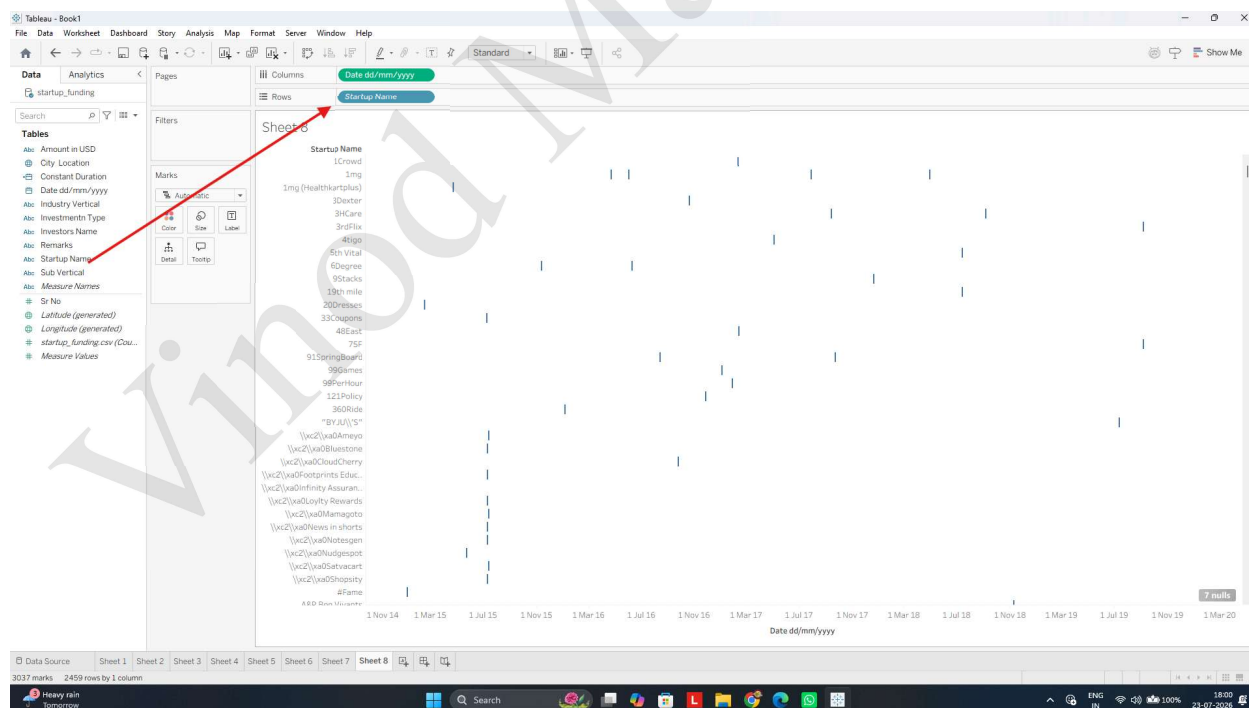
7 marks 1 row by 7 columns

Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3



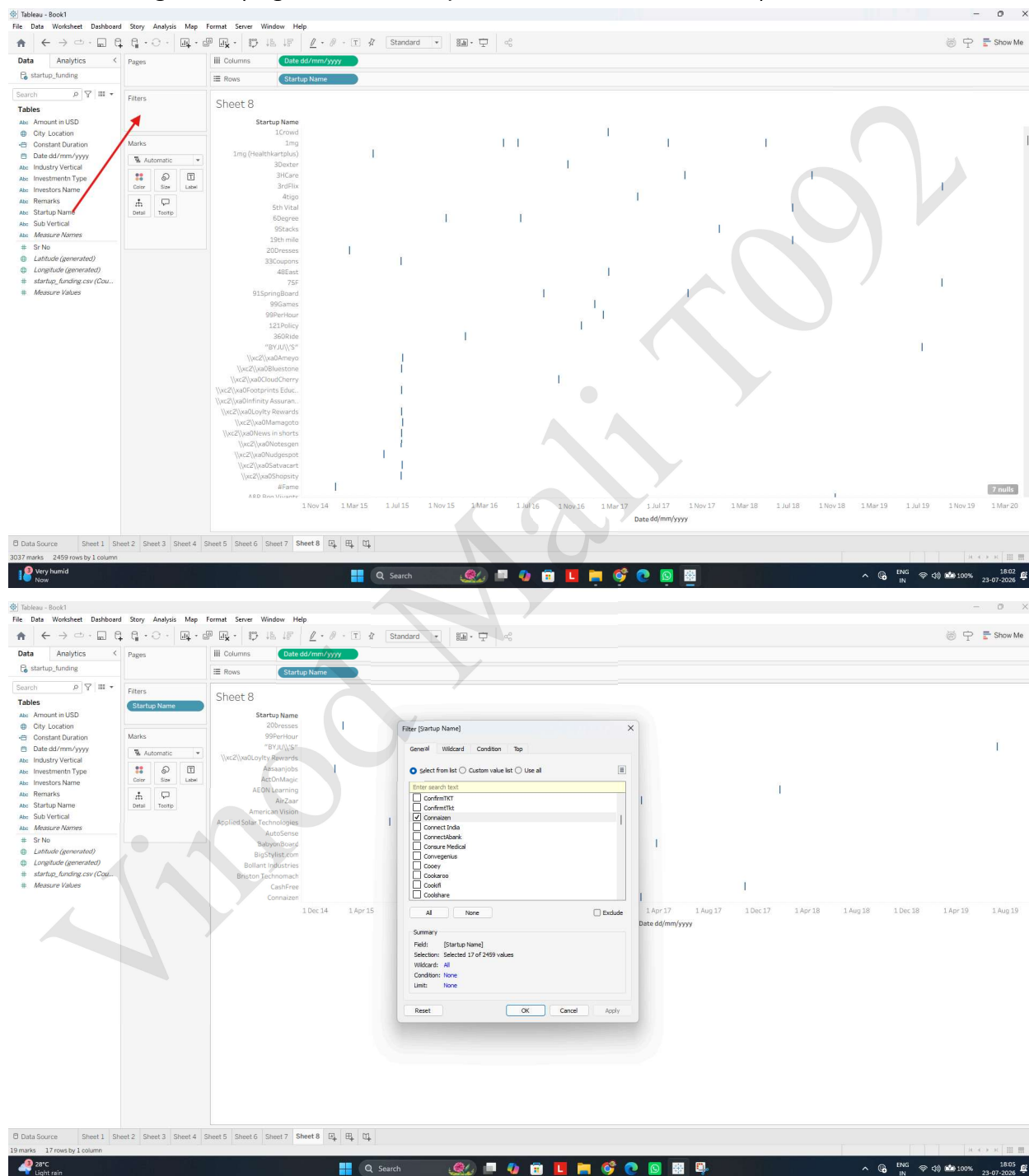
5. Drag your Startup Name dimension from the Data Pane and drop it onto the Rows shelf.



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

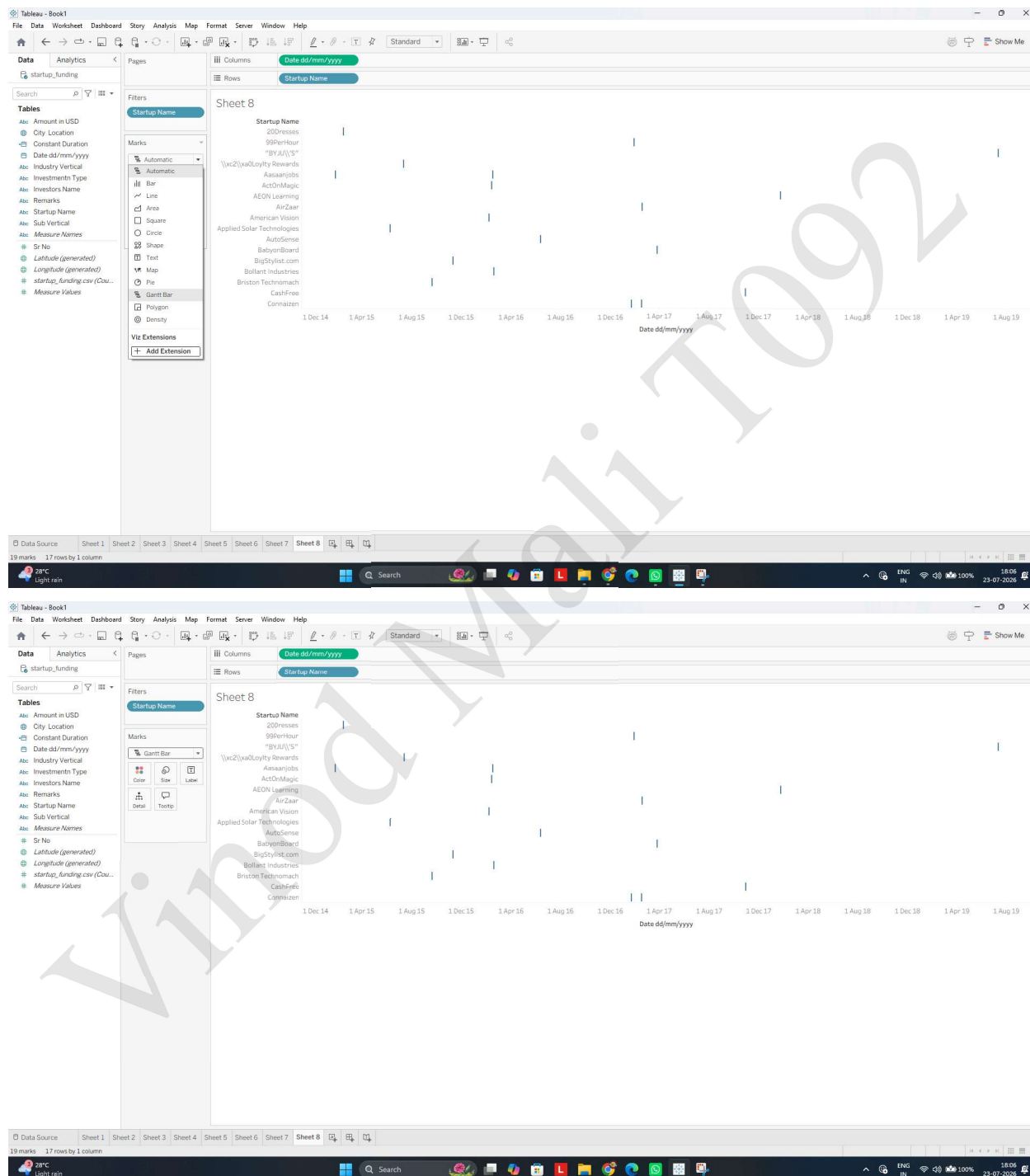
6. Apply the Filter: Because there are thousands of startups, let's look at a focused subset. Drag Startup Name into your Filters shelf, select a few notable Indian tech giants (e.g., BYJU'S, Flipkart, Ola Cabs, Zomato), and hit OK.



Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

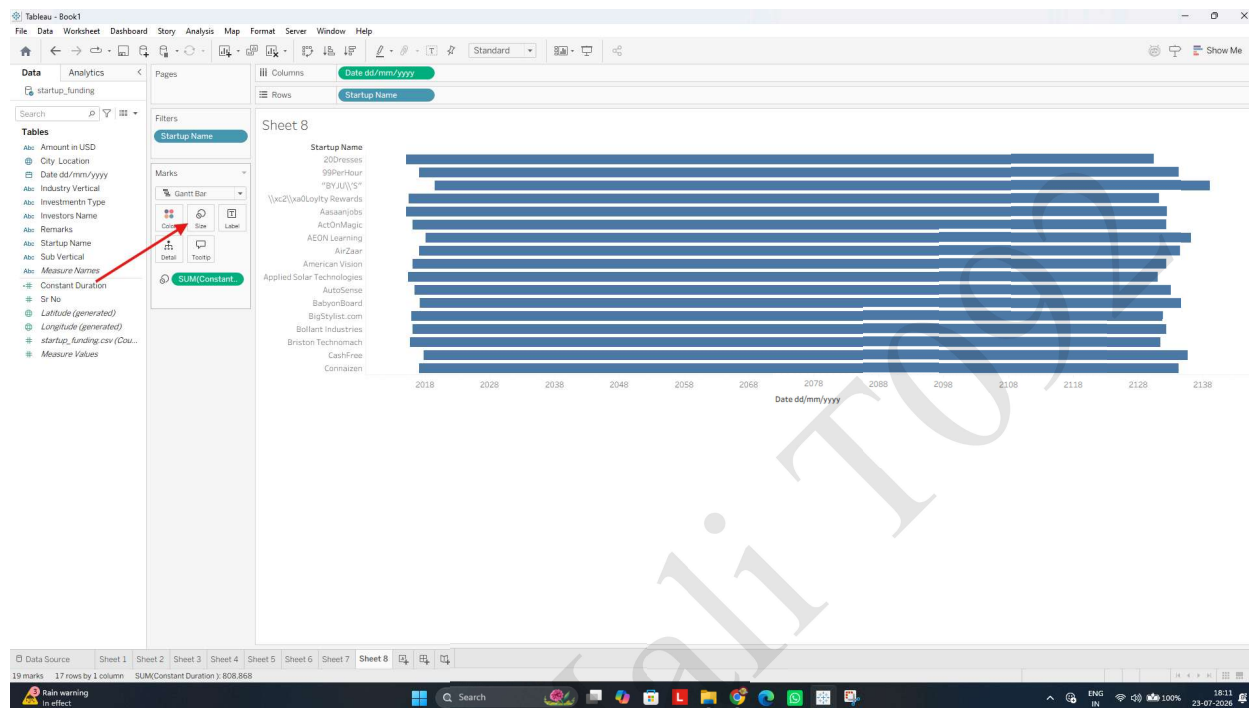
7. Look at your Marks Card dropdown menu (which currently says Automatic). Click it and manually switch the mark style option to Gantt Bar.



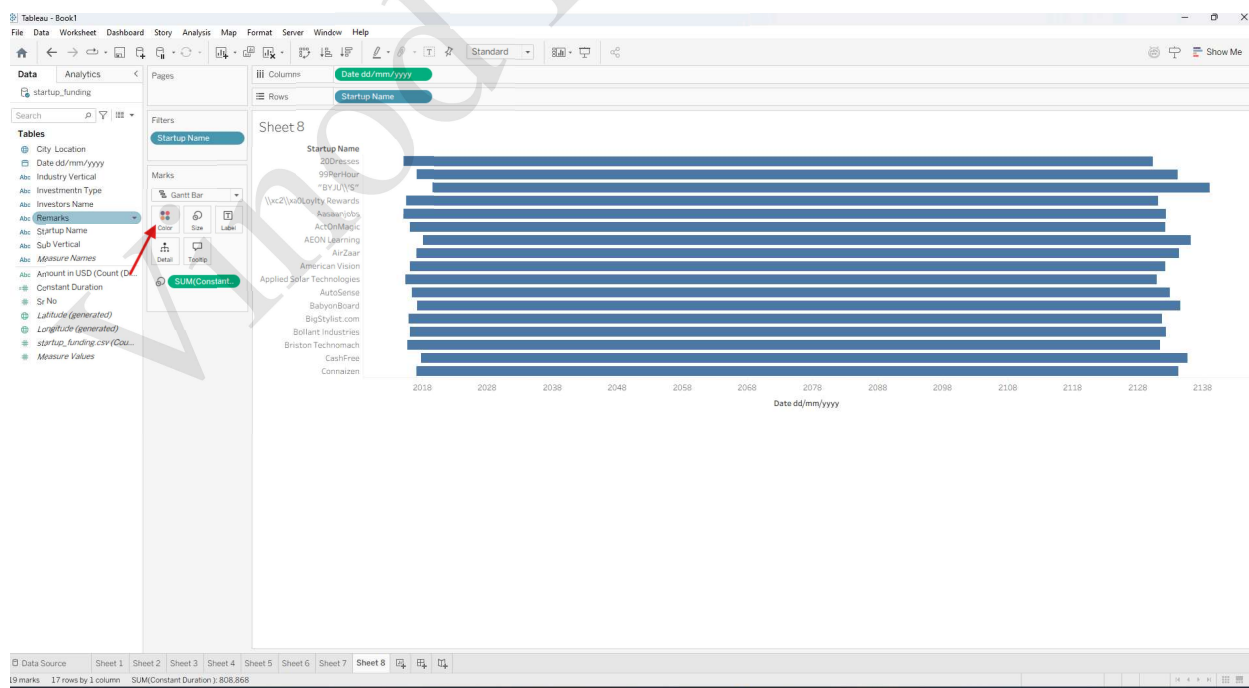
Name :- Vinod Manohar Mali
Roll no :- T092

Sheth L.U.J. & Sir M.V. College Of Science
Subject :- Data Visualization with Power BI and Tableau
Practical no 3

8. Take your newly created calculated field Constant Duration from the Data Pane and drag it directly onto the Size tile inside your Marks card.



9. To make the timeline highly informative, drag your quantitative measure Amount in USD from the Data Pane and drop it directly onto the Color tile in the Marks card.



Name :- Vinod Manohar Mali
Roll no :- T092